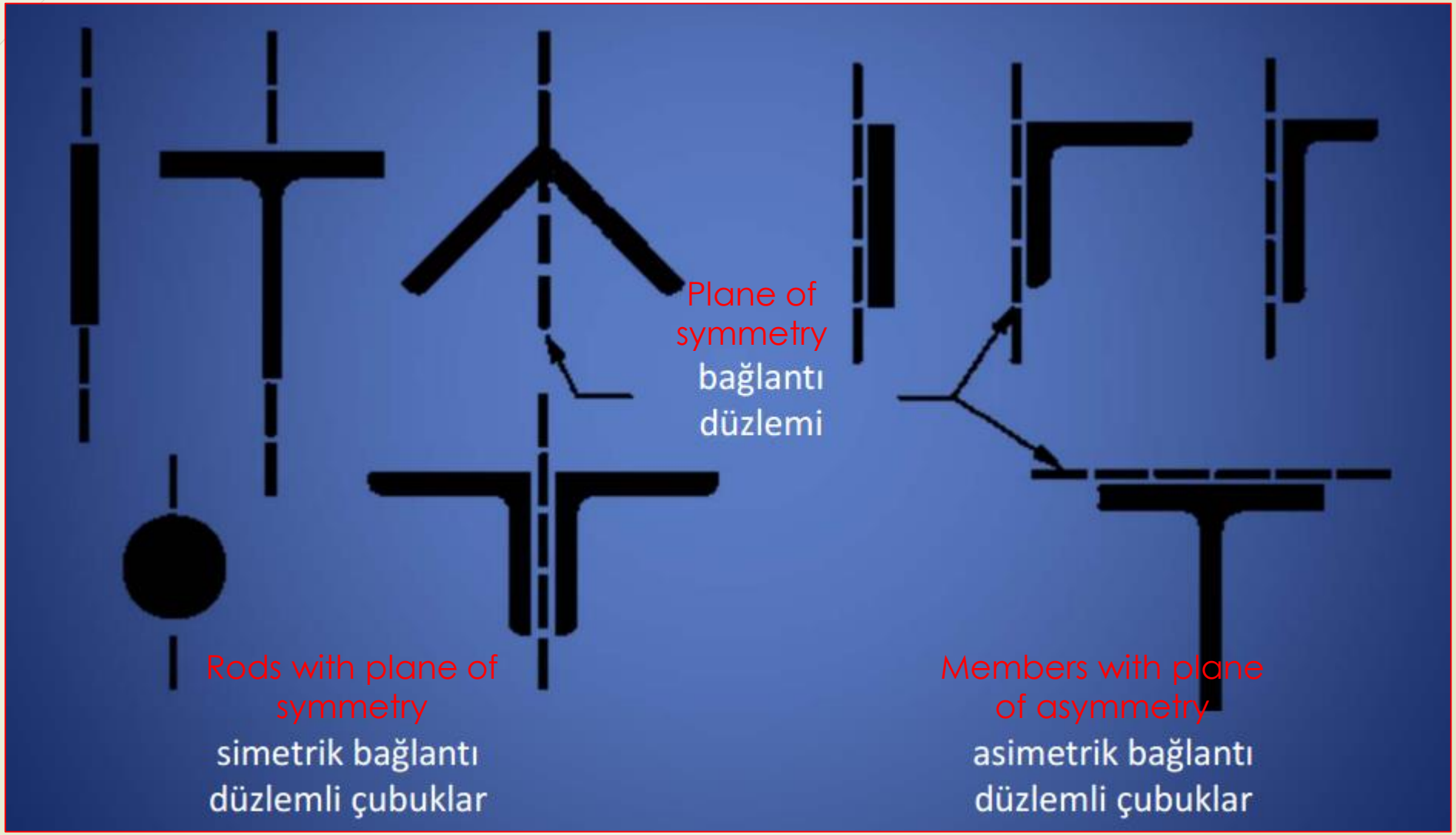
Çekme Elemanları /Tension Members

- Kesit tesiri olarak yalnız eksenleri doğrultusunda ve çekme kuvveti taşıyan elemanlara Çekme Elemanları denir. *(Members that carry tensile force only in the direction of their axes as cross-section effects are called Tensile Members.)*

Çekme elemanları 4 (dört) ana guruba ayrılabilir: *(Tension Members can be orginized in 4 (four) main groups)*

- Teller, halatlar, ipler ve kablolar *(wire, rope, strands, and cables)*
- Çubuklar *(rods or bars)*
- Tek parçalı çekme elemanları *(Single tension members)*
- Çok parçalı çekme elemanları *(Latticed tension members)*

Çekme Elemanları /Tension Members



Teller, ipler, halatlar ve kablolar *(wire, strands, rope and cables)*

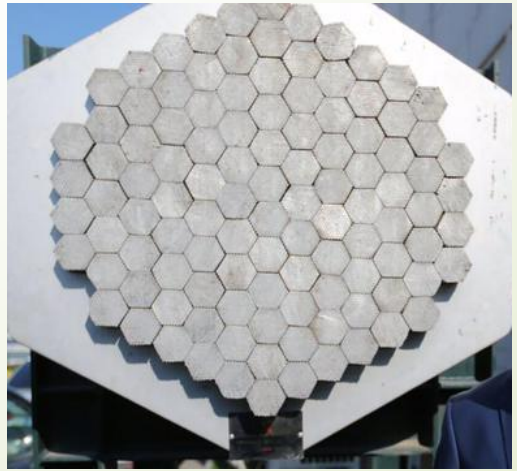
- Tel: Yüksek karbonlu çelik hadde çubuklarından soğuk çekme ile üretilirler. *(Wire: They are produced from high carbon steel rolling bars by cold forming)*
- İp: Bir grup telin merkezi bir telin çevresine simetrik olacak şekilde helezonik olarak yerleştirilmesiyle elde edilir. *(Strand: It is obtained by placing a group of wires spirally symmetrically around a wire.)*
- Halat: Bir çok ipin fiber çekirdek yada merkezi bir ipin çevresinde helezonik olarak yerleştirilmesi ile elde edilir. *(Rope : It is obtained by placing many strands spirally around a fiber core or a central strand.)*
- Kablo: Bir yada daha fazla ip yada halatın birleşmesi ile meydana gelir. *(Cable: It occurs by combining one or more strands or ropes.)*

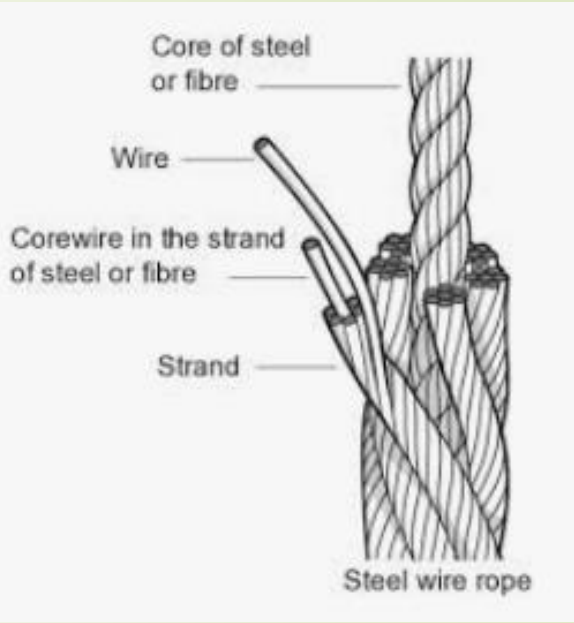


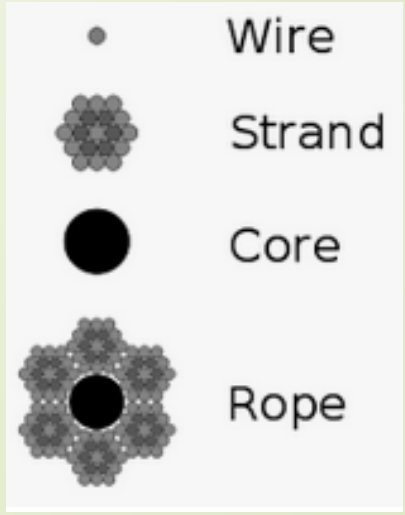
Compacted Wire Rope
Smoother & flatter, tighter strands



Standard Wire Rope
Rounded strands, more open construction





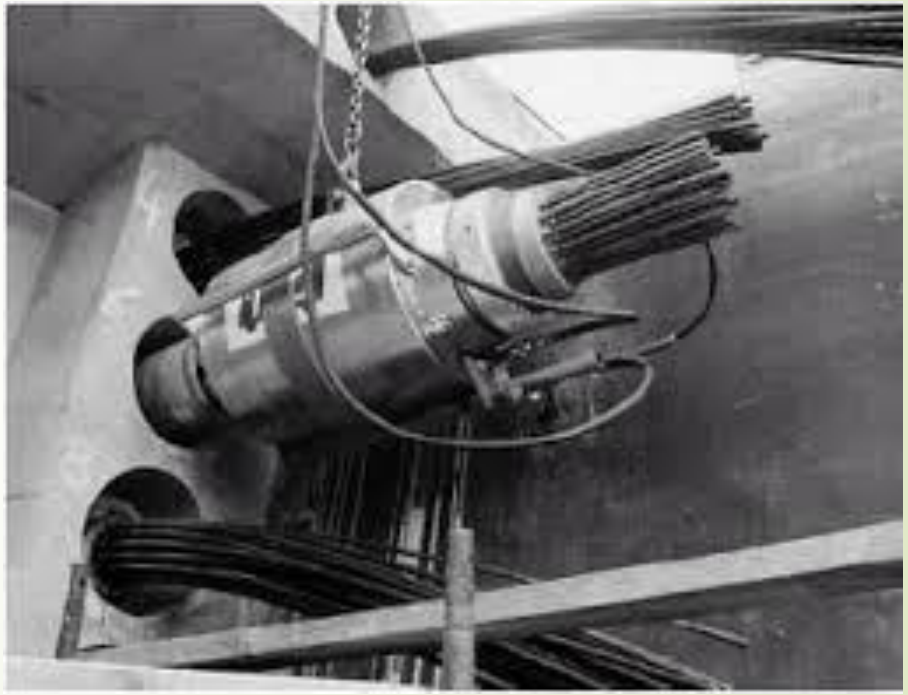


Teller, ipler, halatlar ve kablolar *(wire, strands, rope and cables)*

- Soğukta çekilmiş tellerden meydana gelirler. *(They consist of cold rolled wires)*
- Standart bir akma dayanımına sahip değillerdir. *(They do not have a standard yield strength)*
- Kopma dayanımları dikkate alınarak tasarımı yapılır. *(It is designed considering the ultimate strength)*
- Çekme mukavemeti normal (S235) çeliğin 4-6 katı olmaktadır. Fiyat olarak 2 katıdır. *(Tensile strength is 4-6 times of normal (S235) steel. It is 2 times expensive)*

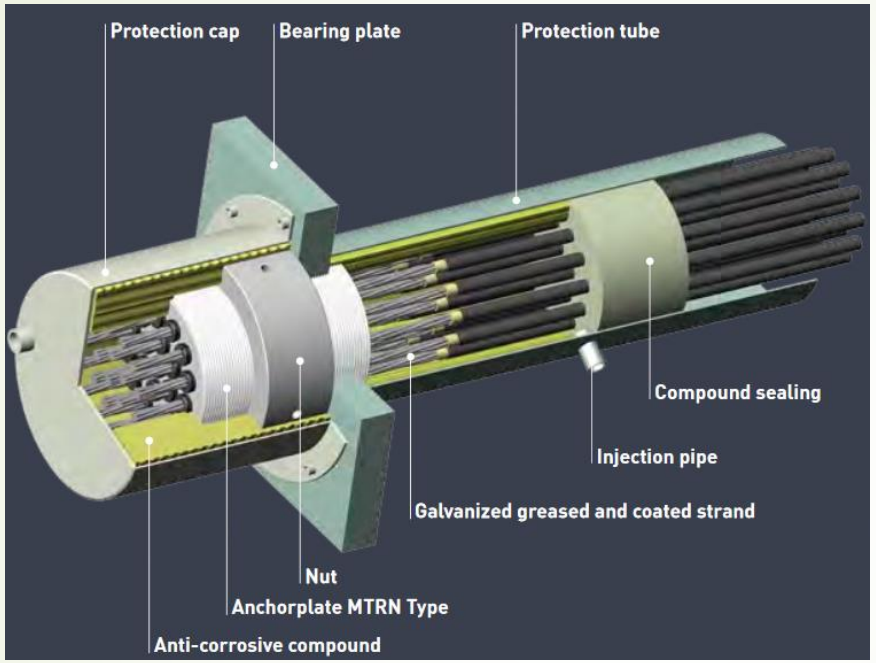
Dezavantajları *(Disadvantages)*

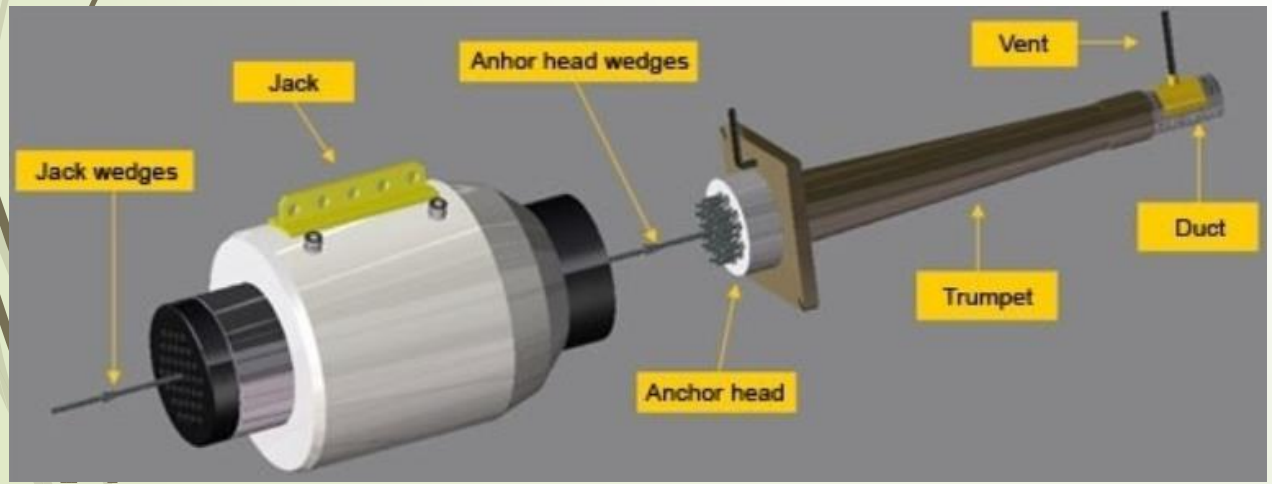
- Basınca dayanamazlar. *(They can't stand the compression)*
- Özel birleşim elemanı gerektirir. *(Requires special join element)*
- Elastisite modülleri düşüktür. (halat ~18000 kN/cm^2 , İp 14000 kN/cm^2) *(Elasticity modules are low) (rope ~18000 kN/cm^2 , stand 14000 kN/cm^2)*

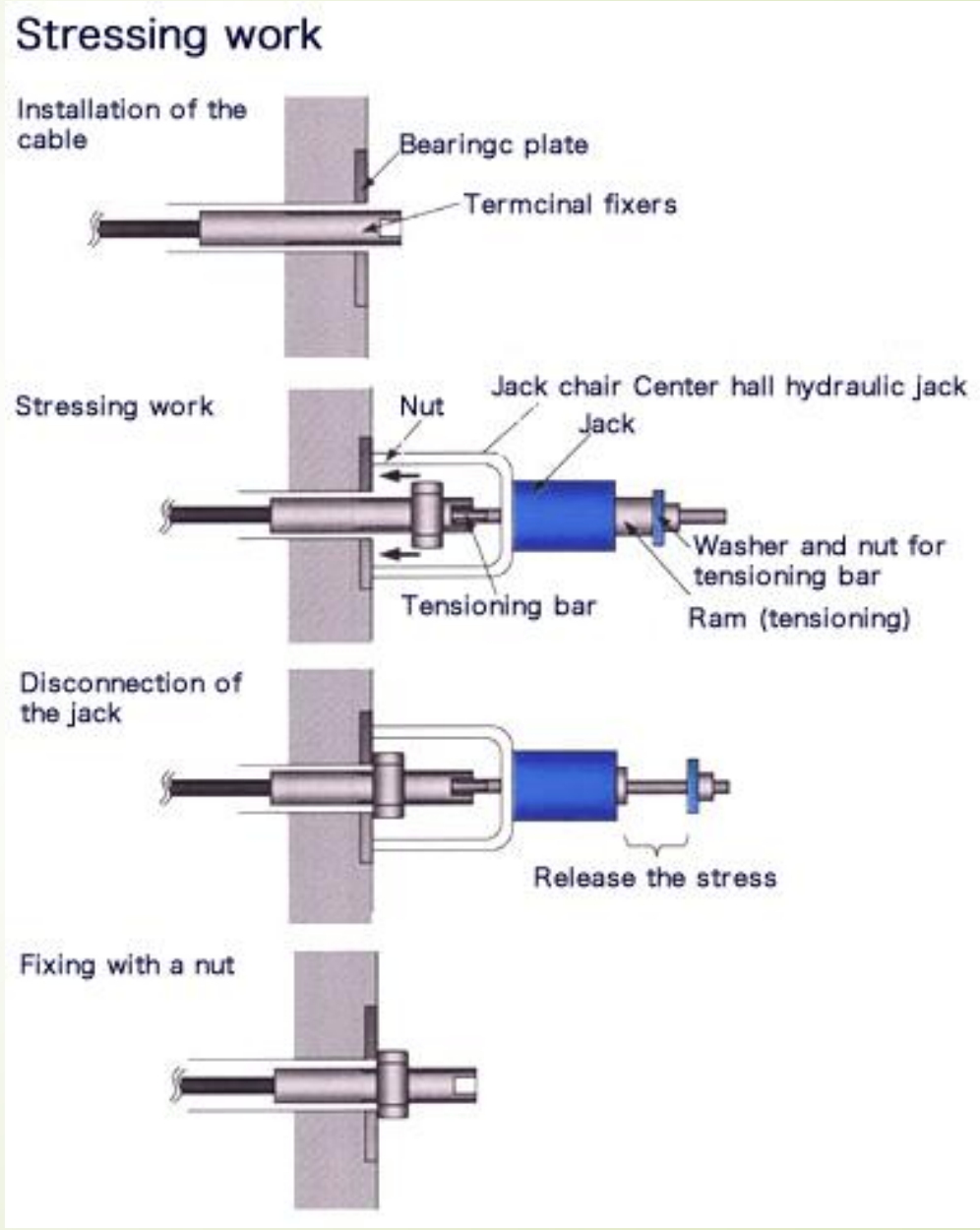


Teller, ipler, halatlar ve kablolar *(wire, strands, rope and cables)*









Çubuklar *(Rods or bars)*

- Sıcak haddeleme ile elde edilen daire veya kare kesitli çubuklardır. *(They are rods or bars obtained by hot rolling)*

Dezavantajları *(Disadvantages)*

- Yeterli rijitliğe sahip olmadıklarından kendi ağırlıkları altında eğilirler *(They sagged under their own weight because they do not have sufficient rigidity)*
- Çok narin oldukları için basınç dayanımları ihmal edilecek kadar azdır. *(Because they are very selender, their compressive strength is negligible)*
- Çubukların istenen uzunlukta üretimleri çok zordur. *(It is very difficult to produce rods or bars in the desired length.)*
- Ana elemanlara bağlantıları sorun olabilir. *(Connections to the main members may be a problem)*





Tek parçalı çekme elemanları */Single Tension Members*

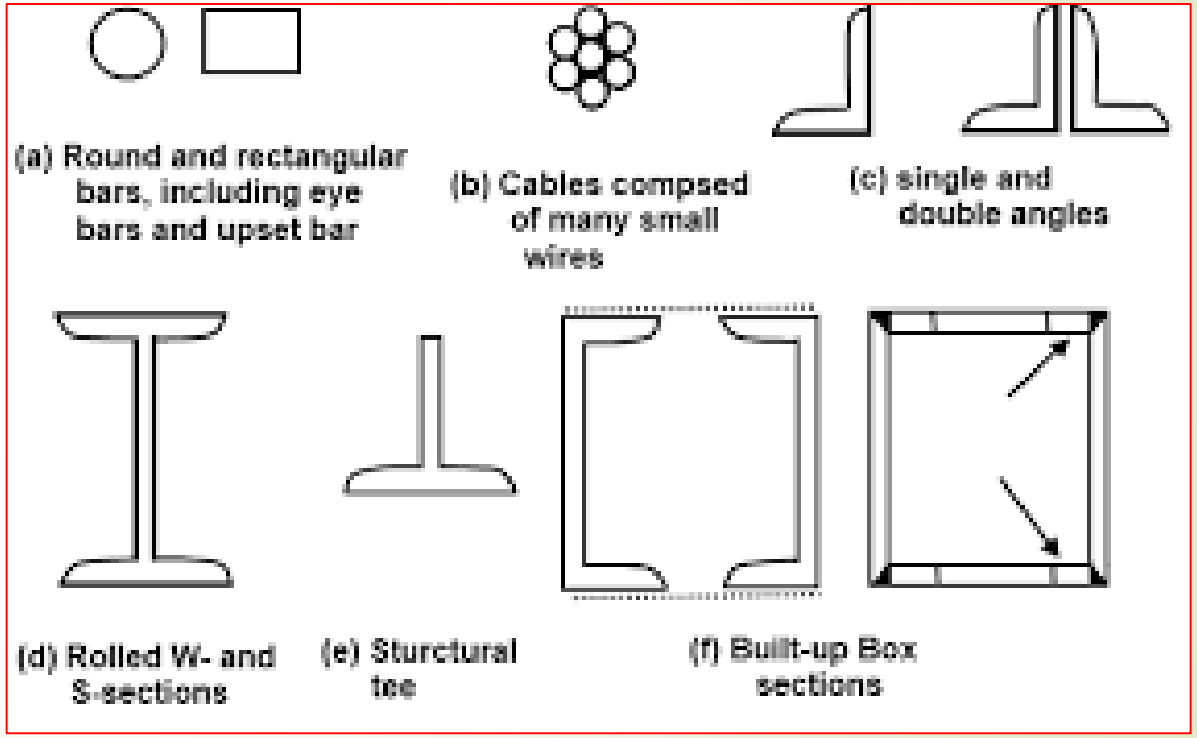
- Hadde profiller kullanılır. Korniyer, I veya U en çok kullanılan kesitlerdir. *(Hot-rolled profiles are used. Cornier, I or U are the most used sections.)*

Dezavantajları *(Disadvantages)*

- Daha rijittirler. *(They are more rigid)*
- Basınç dayanımları vardır. *(They have compressive strength.)*
- Bağlantıları kolaydır. *(They are easy to connect.)*

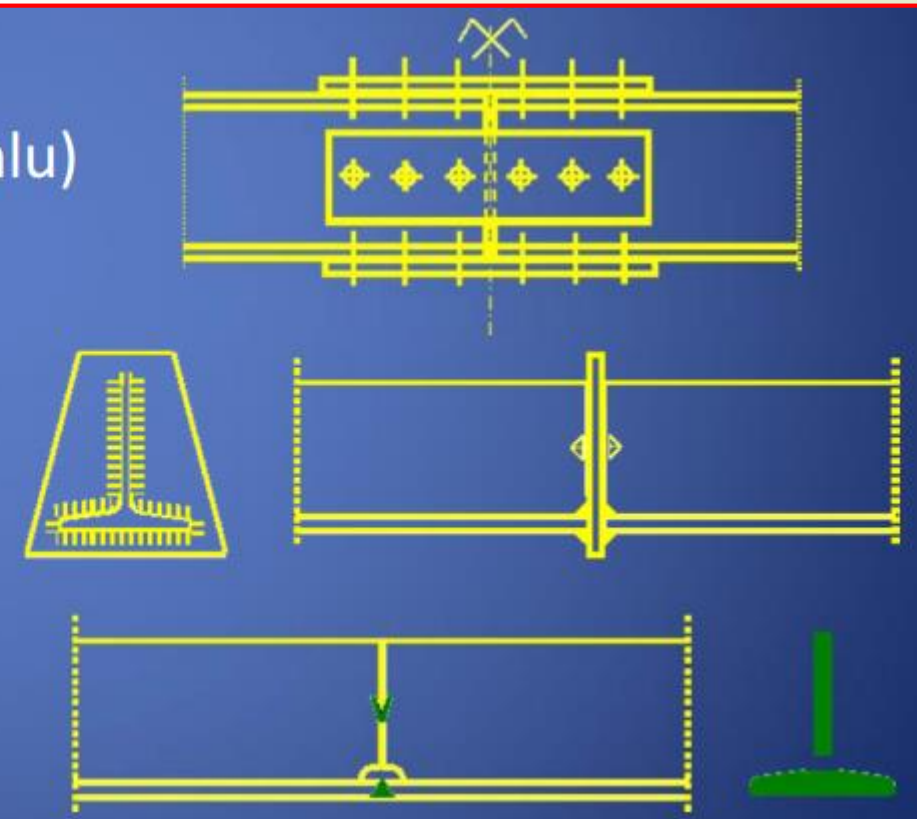
Çok parçalı çekme elemanları /Latticed Tension Members

İki yada daha fazla hadde profilinin uygun şekilde birleştirilmesi ile oluşan elemanlardır. *(They are members formed by combining two or more hot-rolled profiles appropriately)*

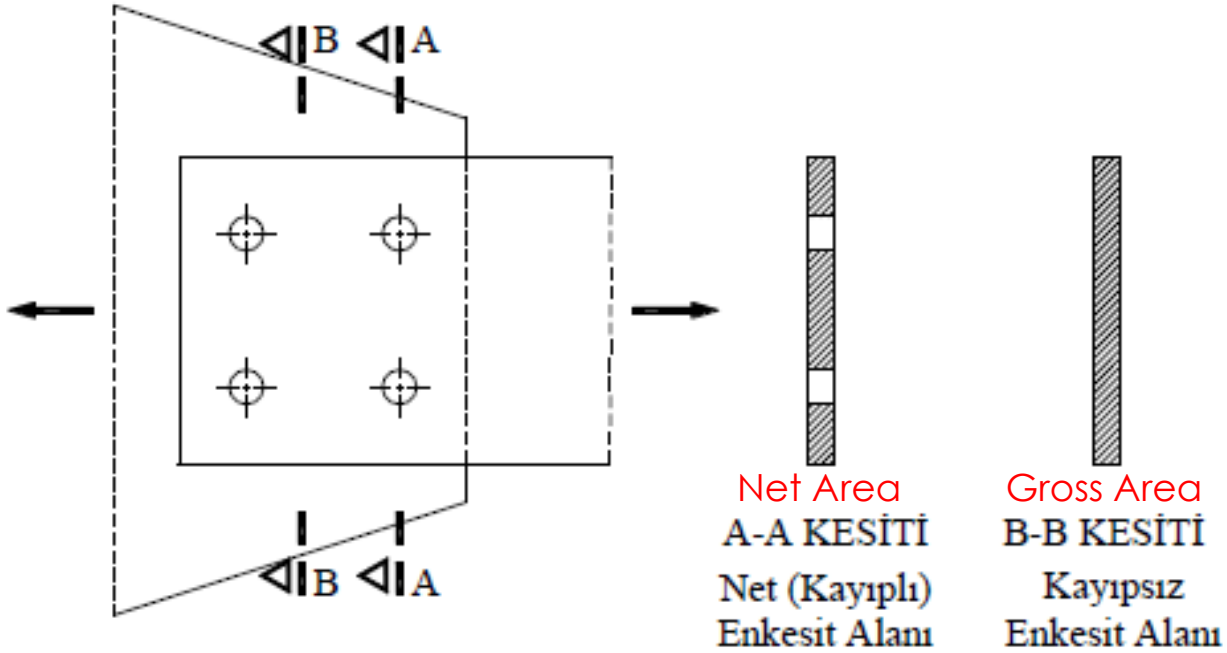


Çekme çubuk ekleri /Splices In Tension Member

- Lamalı ek (kaynaklı veya bulonlu)
- Enleme levhalı ek (kaynaklı)
- Küt ek (kaynaklı)



Toplam alan ve net alan */Gross Area and Net Area*



where

A_e = effective net area, in.^{2} (mm^2)

A_g = gross area of member, in.^{2} (mm^2)

F_y = specified minimum yield stress, ksi (MPa)

F_u = specified minimum tensile strength, ksi (MPa)

Where connections use plug, slot or fillet welds in holes or slots, the effective net area through the holes shall be used in Equation D2-2.

Toplam alan ve net alan */Gross Area and Net Area*

D3. EFFECTIVE NET AREA

The gross area, A_g, and net area, A_n, of tension members shall be determined in accordance with the provisions of Section B4.3.

The effective net area of tension members shall be determined as

$$A_e = A_n U \qquad \text{(D3-1)}$$

where U, the shear lag factor, is determined as shown in Table D3.1.

Toplam alan ve net alan */Gross Area and Net Area*

7.1.2 – Kayıpsız ve Net Enkesit Alanları

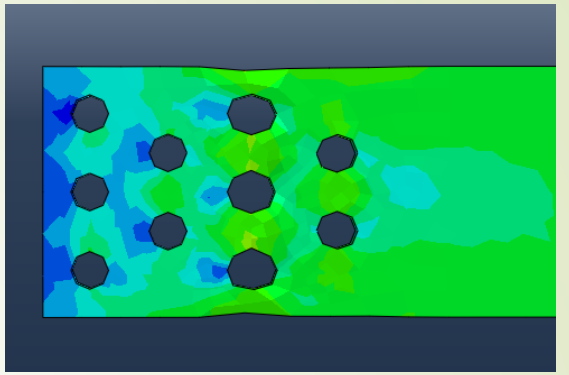
Çekme etkisi altındaki elemanların bu bölümde tanımlanacak olası göçme sınır durumları için, *kayıpsız enkesit alanı*, A_g, ve *net* (*kayıplı*) *enkesit alanı*, A_n, **Bölüm 5.4.3** te verilen ilgili kurallar esas alınarak belirlenecektir.

7.1.3 – Etkin Net Enkesit Alanı

Çekme elemanının birleşim bölgesine tüm enkesit parçalarıyla bulonlu veya kaynaklı olarak bağlanmadığı durumda, gerilme yayılışındaki düzensizliğin hesaba katıldığı *etkin net enkesit alanı*, A_e, **Denk.(7.1)** ile hesaplanacaktır.

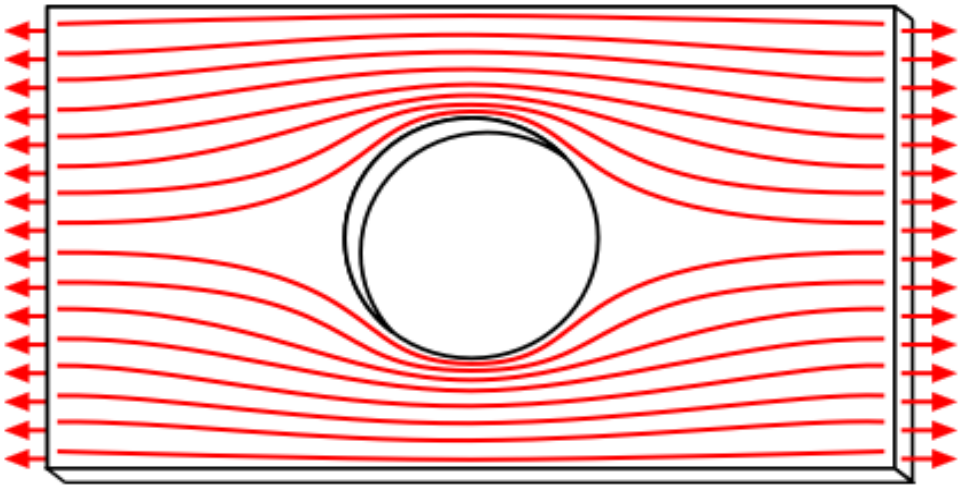
$$A_e = UA_n \qquad (7.1)$$

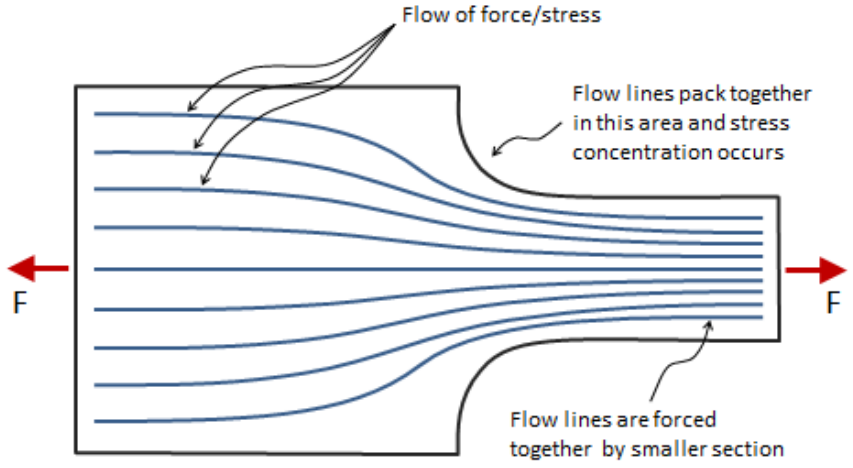
Shear Lag Factor, U
(Düzensiz gerilme dağılımı)



Stress Concentrations- approximate stress state

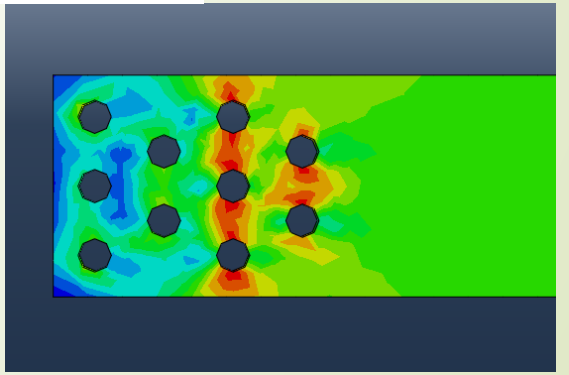
Stress concentrations describe the stress state at abrupt changes in geometry, where the stress profile is non-uniform. Analogous to lines of pressure in (laminar) fluid flow around an immersed body, lines of force 'flow' through geometry, becoming concentrated around the holes (where no material exists to transmit force).





A stress concentration factor (K_s) is applied to the nominal stress to calculate the maximum stress, where $\sigma_{max} = K_s\sigma_n$. Stress concentration factors depend on geometry and are determined analytically, or by experimental data. From the analytical solution of an infinite plate with a single hole, loaded uniaxially, $K_s = 3$. More applicable, from Peterson's Stress Concentration Factors, an infinite plate with a linear hole pattern:

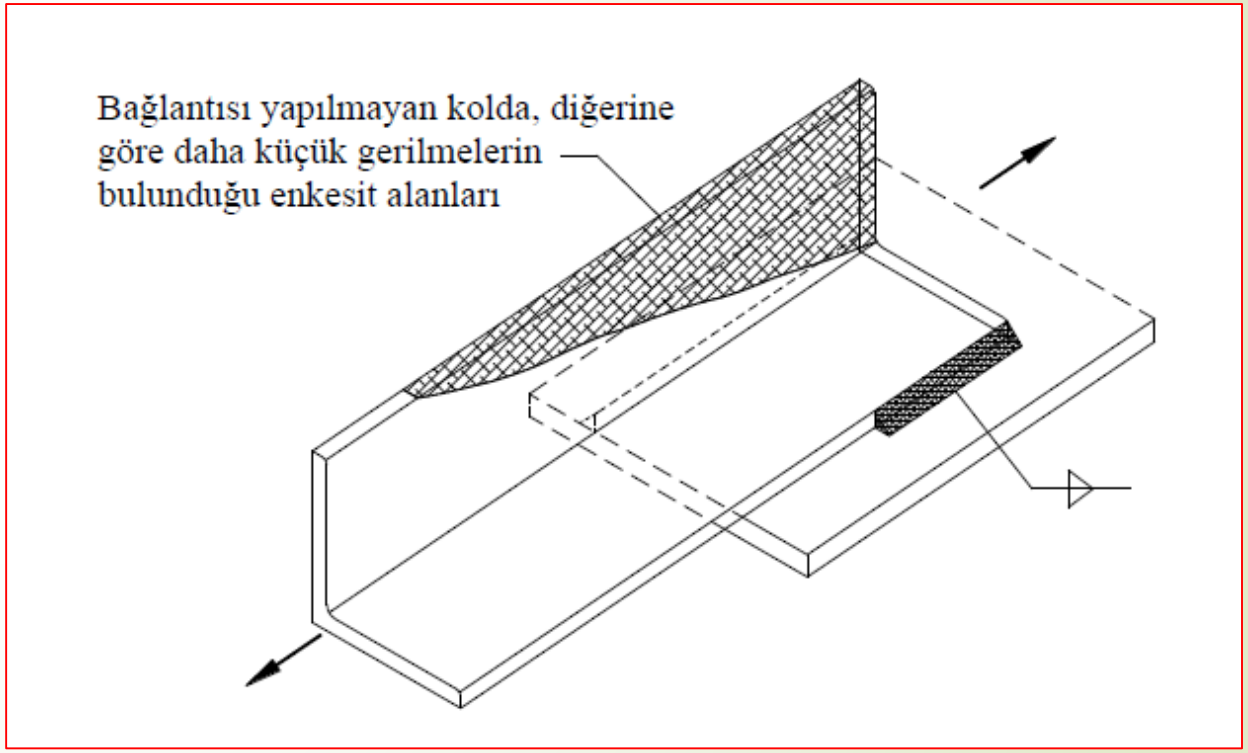
$$\therefore \sigma_{SC} \approx 3\sigma_n$$



Shear Lag Factor, U
(Düzensiz gerilme dağılımı)

- This creates a non-uniform stress distribution around the connection region. That is, the stress trajectory around the connection causes the efficiency of the connection area to be reduced.

- This phenomenon is referred to as shear lag. Shear lag is a complex phenomenon that requires either long calculations or computer modeling to capture its effect.



One of these cases applies to almost all tension members:

$$U_2 = 1 - \frac{\bar{x}_{con}}{\ell_{con}}$$

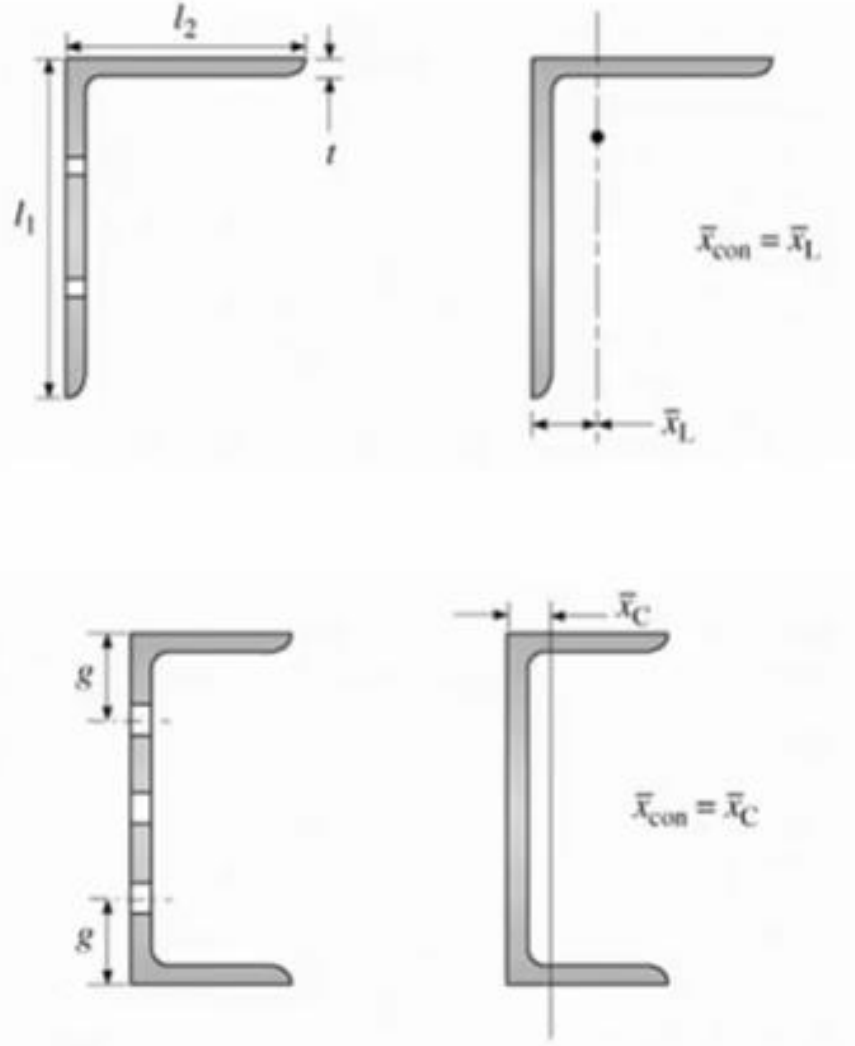
– $\bar{x}_{con}$ = connection eccentricity
  - This is defined as the distance from the connected face of a member to its centroid.

ℓ_{con} = connection length

Examples of connection eccentricity:

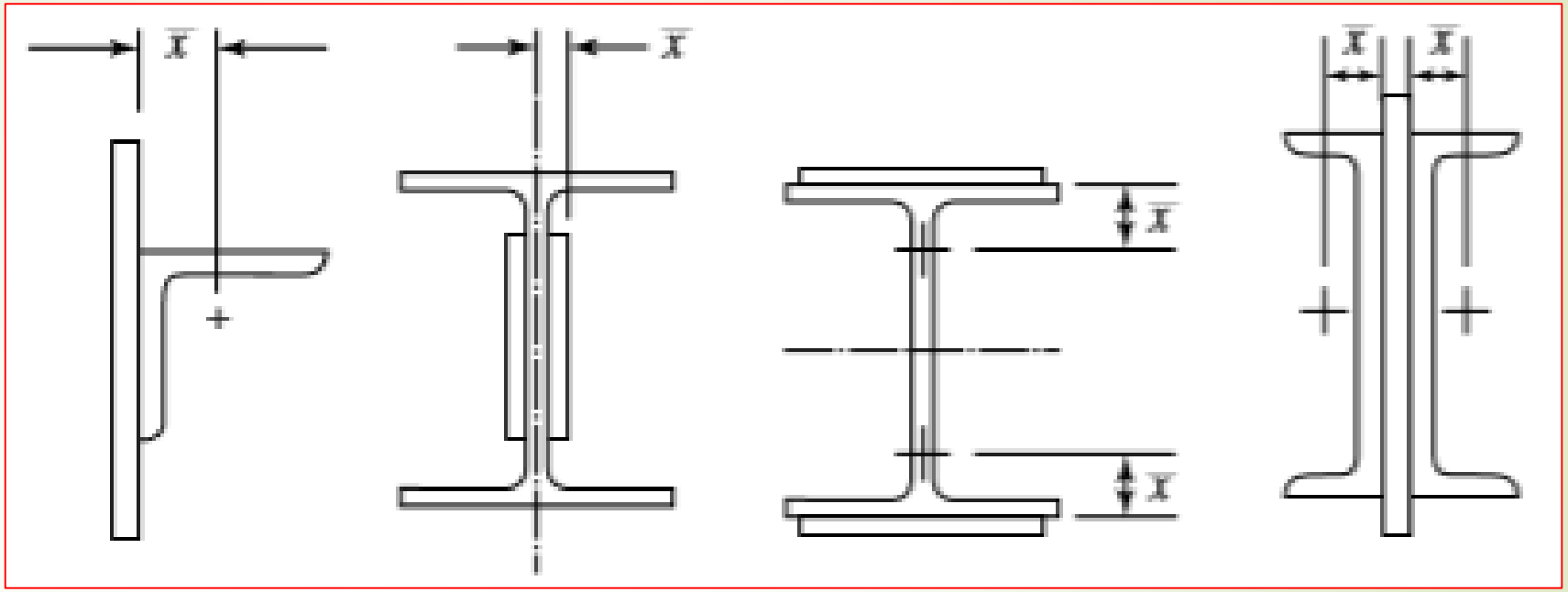
– For most cases, you can simply look up the appropriate centroid value.

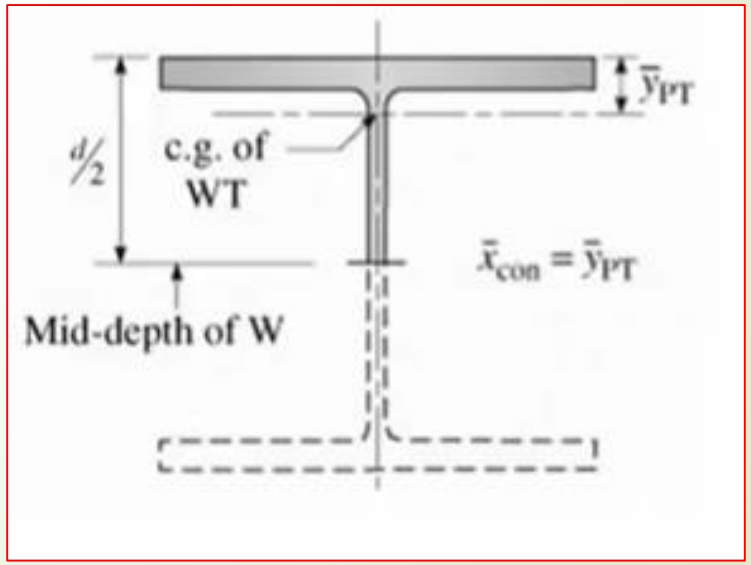
– The major concern is that you're using the correct value!



Shear Lag Factor, U

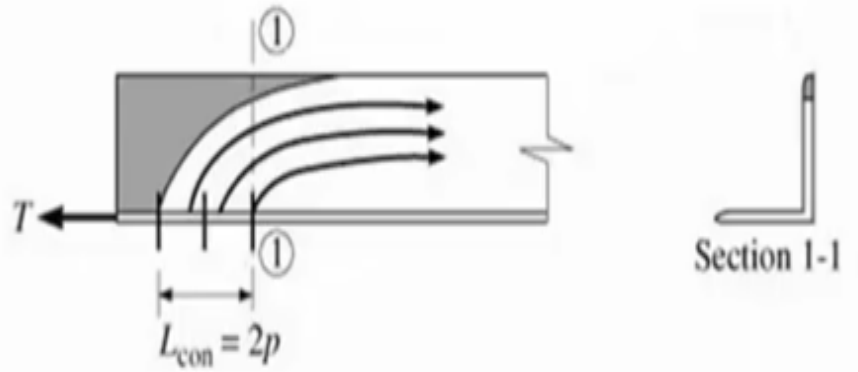
(Düzensiz gerilme dağılımı)

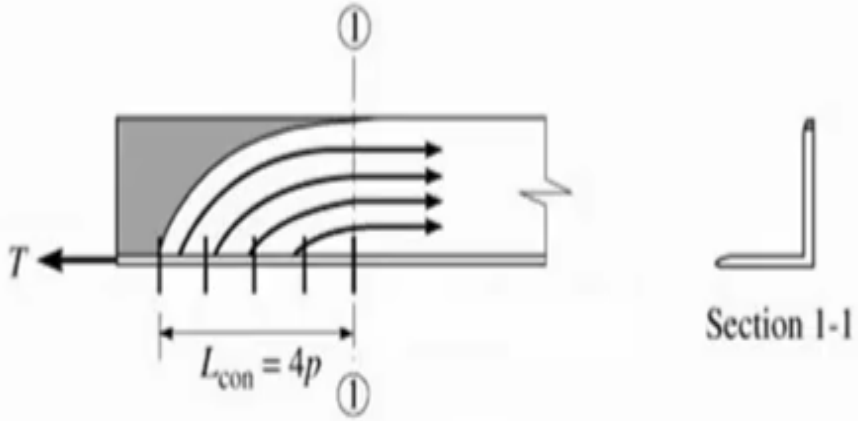




The effect of connection length can be shown as follows:

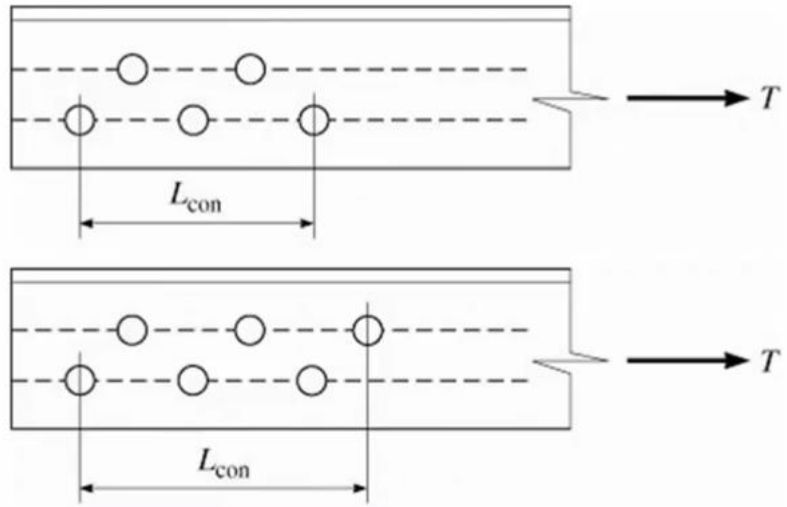
– As the connection gets longer, the stress has more room to propagate the cross-section





For bolted connections:

– Connection length is taken as the out-to-out distance between bolt holes:



For welded connections:

– Connection length is taken as the average longitudinal weld length:

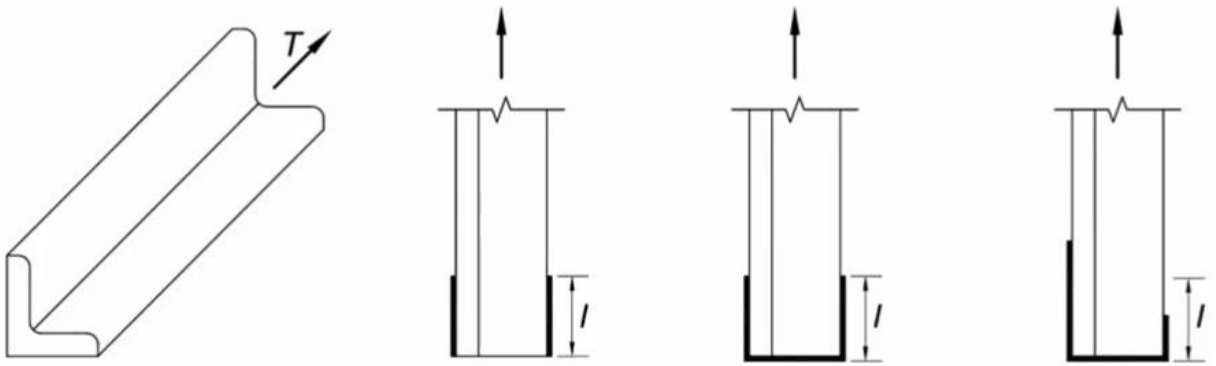
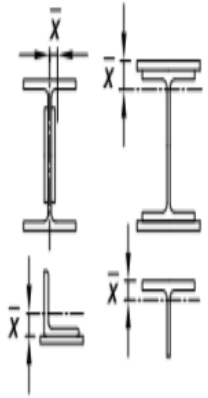
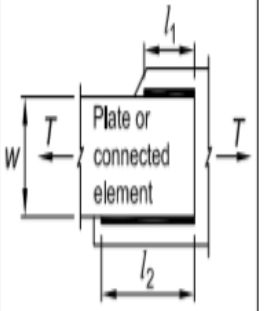
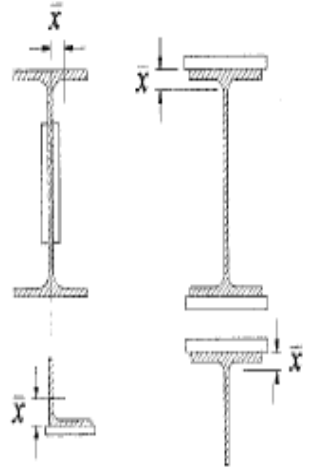
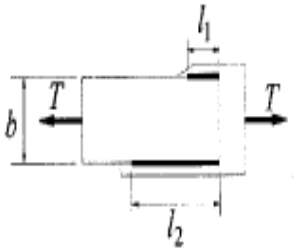
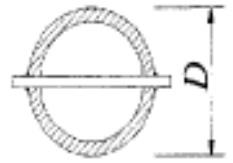
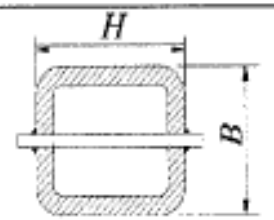
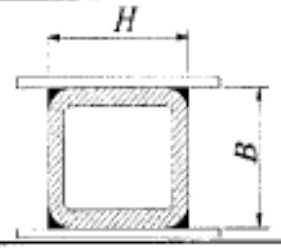


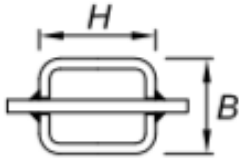
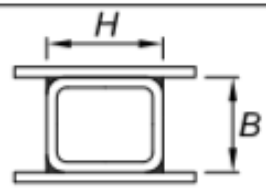
TABLE D3.1
Shear Lag Factors for Connections to Tension Members

Case	Description of Element	Shear Lag Factor, U	Example
1	All tension members where the tension load is transmitted directly to each of the cross-sectional elements by fasteners or welds (except as in Cases 4, 5 and 6).	$U = 1.0$	–
2	All tension members, except HSS, where the tension load is transmitted to some but not all of the cross-sectional elements by fasteners or by longitudinal welds in combination with transverse welds. Alternatively, Case 7 is permitted for W, M, S and HP shapes. (For angles, Case 8 is permitted to be used.)	$U = 1 - \frac{\bar{x}}{l}$	
3	All tension members where the tension load is transmitted only by transverse welds to some but not all of the cross-sectional elements.	$U = 1.0$ and A_n = area of the directly connected elements	–
4[a]	Plates, angles, channels with welds at heels, tees, and W-shapes with connected elements, where the tension load is transmitted by longitudinal welds only. See Case 2 for definition of $\bar{x}$.	$U = \frac{3l^2}{3l^2 + w^2}\left(1 - \frac{\bar{x}}{l}\right)$	

TABLO - 7.1 GERİLME DÜZENSİZLİĞİ ETKİ KATSAYISI

Durum	Çekme Elemanı Tanımı	Gerilme Düzensizliği Etki Katsayısı, U	Örnek
1	Çekme elemanlarının birleşim bölgesine tüm enkesit parçalarıyla bağlanması durumu	$U = 1.0$	–
2	I, U ve T- enkesitli çekme elemanlarının tüm enkesit parçalarıyla bağlanmaması durumu	$U = 1 - \frac{\bar{x}}{l}$	
3	Kuvvetin çekme doğrultusuna dik kaynaklarla aktarıldığı birleşim	$U = 1.0$ A_n : doğrudan birleşen enkesit parçasının alanı	–
4[a]	Çekme kuvvetinin sadece boyuna doğrultudaki kaynaklarla aktarıldığı levhalar, korniyerler, U-profiller, T-enkesitli elemanlar ve birleşim elemanları ile bağlanan I-profillerin birleşimi ($\bar{x}$ nin tanımı için Durum 2 ye bakınız).	$U = \frac{3l^2}{\left(3l^2 + b^2\right)}\left(1 - \frac{\bar{x}}{l}\right)$	

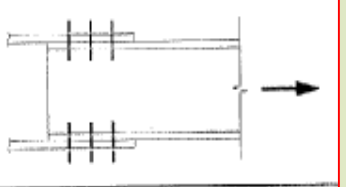
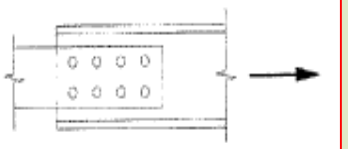
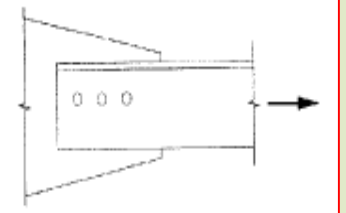
5	Bir birleşim levhasına bağlanan boru enkesitli profiller[b]		$l \geq 1.3D \rightarrow U = 1.0$ $D \leq l < 1.3D$ $\rightarrow U = 1 - \bar{x}/l$ $\bar{x} = D/\pi$	
6	Birleşim levhasına bağlanan kutu enkesitli profiller[b]	Bir adet merkezi birleşim levhası kullanılması durumu	$l \geq H \rightarrow U = 1 - \bar{x}/l$ $\bar{x} = \frac{B^2 + 2BH}{4(B+H)}$	
		Yan kenarlarda iki adet birleşim levhası kullanılması durumu	$l \geq H \rightarrow U = 1 - \bar{x}/l$ $\bar{x} = \frac{B^2}{4(B+H)}$	

5	Round HSS with a single concentric gusset plate through slots in the HSS.		$l \geq 1.3D,\ U = 1.0$ $D \leq l < 1.3D,\ U = 1 - \frac{\bar{x}}{l}$ $\bar{x} = \frac{D}{\pi}$	
6	Rectangular HSS.	with a single concentric gusset plate	$l \geq H,\ U = 1 - \frac{\bar{x}}{l}$ $\bar{x} = \frac{B^2 + 2BH}{4(B+H)}$	
		with two side gusset plates	$l \geq H,\ U = 1 - \frac{\bar{x}}{l}$ $\bar{x} = \frac{B^2}{4(B+H)}$	

7	W-, M-, S- or HP-shapes, or tees cut from these shapes. (If U is calculated per Case 2, the larger value is permitted to be used.)	with flange connected with three or more fasteners per line in the direction of loading	$b_f \geq \frac{2}{3}d,\ U = 0.90$ $b_f < \frac{2}{3}d,\ U = 0.85$	–
		with web connected with four or more fasteners per line in the direction of loading	$U = 0.70$	–
8	Single and double angles. (If U is calculated per Case 2, the larger value is permitted to be used.)	with four or more fasteners per line in the direction of loading	$U = 0.80$	–
		with three fasteners per line in the direction of loading (with fewer than three fasteners per line in the direction of loading, use Case 2)	$U = 0.60$	–

B = overall width of rectangular HSS member, measured 90° to the plane of the connection, in. (mm); D = outside diameter of round HSS, in. (mm); H = overall height of rectangular HSS member, measured in the plane of the connection, in. (mm); d = depth of section, in. (mm); for tees, d = depth of the section from which the tee was cut, in. (mm); l = length of connection, in. (mm); w = width of plate, in. (mm); $\bar{x}$ = eccentricity of connection, in. (mm).

[a] $l = \frac{l_1 + l_2}{2}$, where l_1 and l_2 shall not be less than 4 times the weld size.

7	I ve ½ I enkesitli profiller[c]	Başlıkların kuvvet doğrultusunda bir sırada 3 veya daha fazla bulonla bağlandığı birleşim	$b_f / d \geq 2/3 \rightarrow U = 0.90$ $b_f / d < 2/3 \rightarrow U = 0.85$	
		Gövdenin kuvvet doğrultusunda bir sırada 4 veya daha fazla bulonla bağlandığı birleşim	$U = 0.70$	
8	Tek ve çift korniyer[c]	Kuvvet doğrultusunda bir sırada 4 veya daha fazla bulon bulunan birleşim	$U = 0.80$	
		Kuvvet doğrultusunda bir sırada 2 veya 3 bulon bulunan birleşim	$U = 0.60$	

$\bar{x}$: Birleşim etki alanı ağırlık merkezinin birleşim düzlemine dik uzaklığı (dışmerkezlik etkisi)

l : Yük doğrultusundaki etkin birleşim uzunluğu (bulonlu birleşimlerde en uzun bulon sırasındaki en dış bulon merkezleri arasındaki uzunluk)

b_f : Başlık genişliği.

d : Enkesit yüksekliği.

[a] : $l = \frac{l_1 + l_2}{2}$

[b] : Kaynak uzunluğu, l, kutu profilin genişliği, H veya boru profilin çapı, D den küçük olamaz.

[c] : Bulonlu çekme elemanlarının ön boyutlandırılması aşamasında, gerilme düzensizliği etki katsayısı için **Tablo 7.1** de verilen ortalama değerler kullanılabilir. Birleşimin tasarımından sonra, **Tablo 7.1** de verilen ifadeler ile hesaplanan daha büyük U değerlerinin kullanılmasına izin verilmektedir.

Design for Strength Using Load and Resistance Factor Design (LRFD)

Design according to the provisions for load and resistance factor design (LRFD) satisfies the requirements of this Specification when the design strength of each structural component equals or exceeds the required strength determined on the basis of the LRFD load combinations. All provisions of this Specification, except for those in Section B3.2, shall apply.

Design shall be performed in accordance with Equation B3-1:

$$R_u \le \phi R_n \qquad \text{(B3-1)}$$

where

R_u = required strength using LRFD load combinations
R_n = nominal strength
ϕ = resistance factor
ϕR_n = design strength

The nominal strength, R_n, and the resistance factor, ϕ, for the applicable limit states are specified in Chapters D through K.

Design for Strength Using Allowable Strength Design (ASD)

Design according to the provisions for allowable strength design (ASD) satisfies the requirements of this Specification when the allowable strength of each structural component equals or exceeds the required strength determined on the basis of the ASD load combinations. All provisions of this Specification, except those of Section B3.1, shall apply.

Design shall be performed in accordance with Equation B3-2:

$$R_a \le \frac{R_n}{\Omega} \qquad \text{(B3-2)}$$

where

R_a = required strength using ASD load combinations
R_n = nominal strength
Ω = safety factor
R_n/Ω = allowable strength

The nominal strength, R_n, and the safety factor, Ω, for the applicable limit states are specified in Chapters D through K.

Bölüm 7. Eksenel Çekme Kuvveti Etkisi /Design of Members For Tension

7.2 ÇEKME KUVVETİ DAYANIMI

Tasarım çekme kuvveti dayanımı, $\phi_t T_n$, (YDKT) veya *güvenli çekme kuvveti dayanımı*, T_n/Ω_t, (GKT), eksenel çekme kuvveti etkisindeki elemanın, *akma sınır durumu*, *kırılma sınır durumu* ve *blok kırılma sınır durumlarına* göre hesaplanacak dayanımların en küçüğü olarak alınacaktır.

Çekme elemanlarının uç birleşimlerinde blok kırılma sınır durumu **Bölüm 13.4.3** te verilmektedir.

D2. TENSILE STRENGTH

The design tensile strength, $\phi_t P_n$, and the allowable tensile strength, P_n/Ω_t, of tension members shall be the lower value obtained according to the limit states of tensile yielding in the gross section and tensile rupture in the net section.

Akma Sınır Durumu */Limit State of Yielding*

5.2.2 Yük ve Dayanım Katsayıları ile Tasarım (YDKT)

5.2.3 Güvenlik Katsayıları ile Tasarım (GKT)

7.2.1 – Akma Sınır Durumu tensile yielding in the gross section

Çekme elemanlarında akma sınır durumu için karakteristik çekme kuvveti dayanımı, T_n, kayıpsız enkesit alanı kullanılarak **Denk.(7.2)** ile hesaplanacaktır.

$$T_n = F_y A_g \qquad \textbf{(7.2)}$$

Tasarım çekme kuvveti dayanımı, $\phi_t T_n$, (YDKT) veya *güvenli çekme kuvveti dayanımı*, T_n/Ω_t, (GKT),

$$\phi_t = 0.90 \text{ (YDKT)} \qquad \text{veya} \qquad \Omega_t = 1.67 \text{ (GKT)}$$

alınarak belirlenecektir.

Kırılma Sınır Durumu */Limit State of Bearing*

7.2.2 – Kırılma Sınır Durumu

Çekme elemanlarında kırılma sınır durumu için, karakteristik çekme kuvveti dayanımı, T_n, etkin net enkesit alanı kullanılarak **Denk.(7.3)** ile hesaplanacaktır.

$$T_n = F_u A_e \qquad \text{tensile rupture in the net section} \qquad \textbf{(7.3)}$$

Tasarım çekme kuvveti dayanımı, $\phi_t T_n$, (YDKT) veya *güvenli çekme kuvveti dayanımı*, T_n/Ω_t, (GKT),

$$\phi_t = 0.75 \text{ (YDKT)} \qquad \text{veya} \qquad \Omega_t = 2.00 \text{ (GKT)}$$

alınarak belirlenecektir.

Buradaki terimler aşağıda açıklanmıştır.

A_e : Etkin net enkesit alanı.

A_g : Kayıpsız enkesit alanı.

F_y : Yapısal çelik karakteristik akma gerilmesi.

F_u : Yapısal çelik karakteristik çekme dayanımı.

Block Shear Rupture

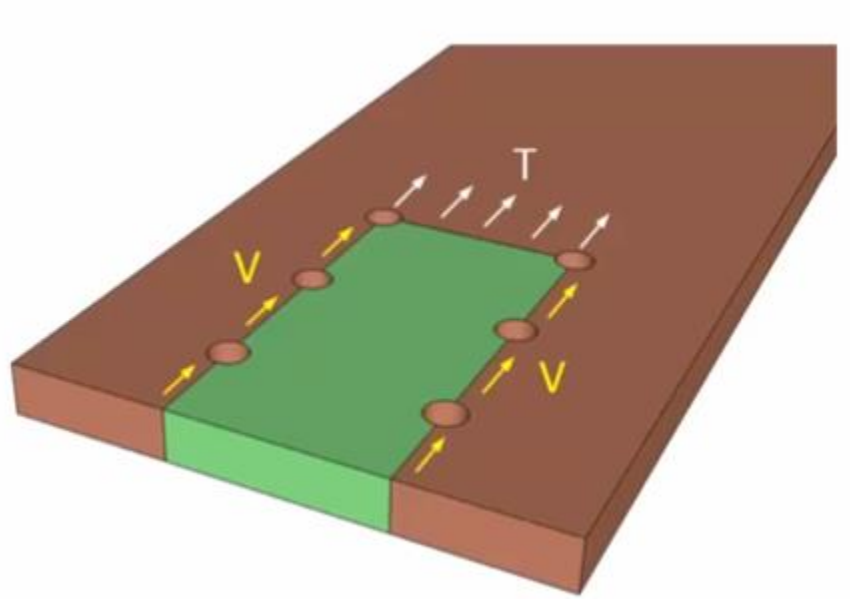
- Block shear rupture is a limit state of
  - Tension fracture along one path
  - Shear yielding or shear fracture along another path.



Test @ Georgia Tech



Test @ WVU



Limit State of Tension and Shear Rupture (Blok kırılma durumu)

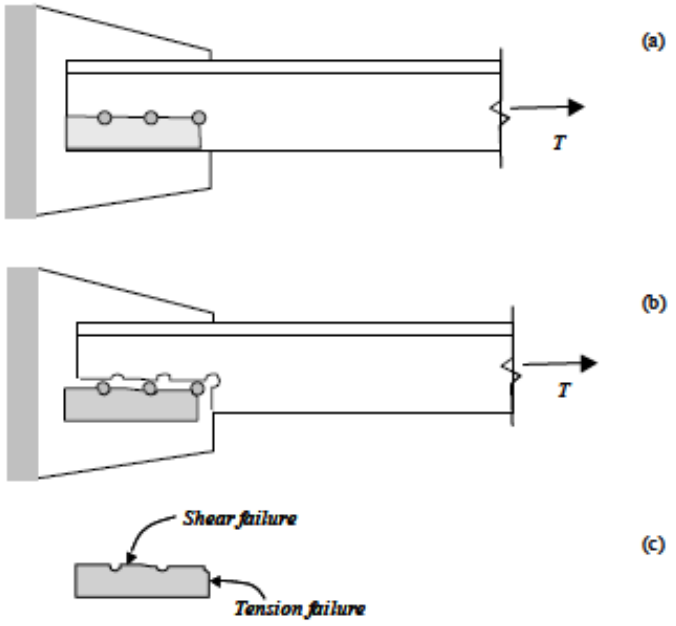
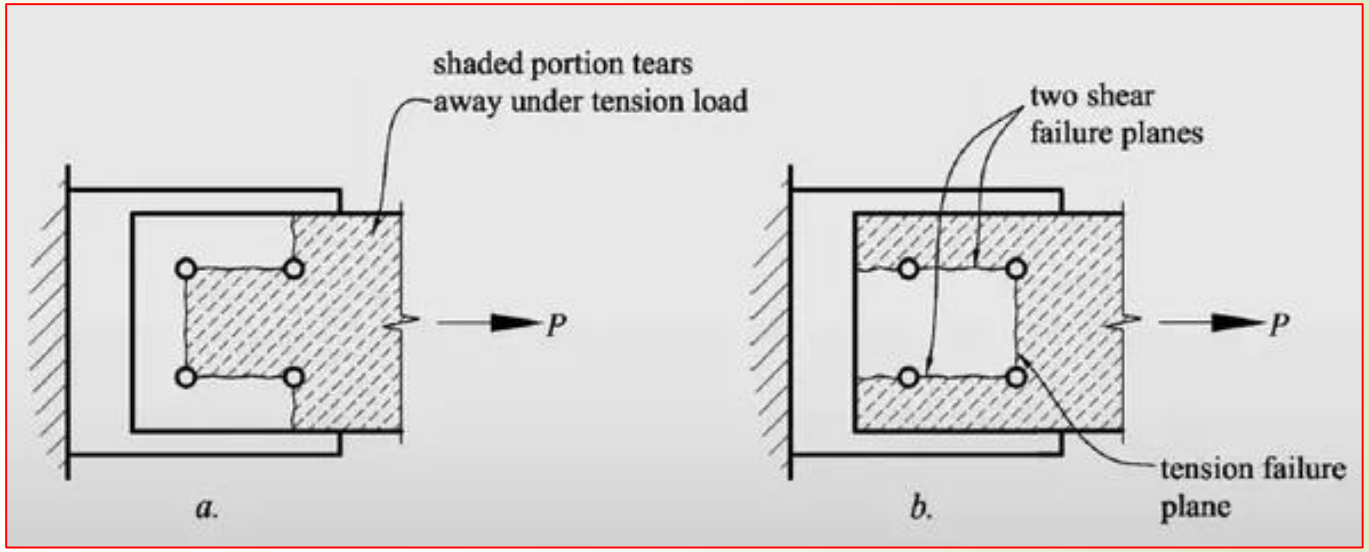
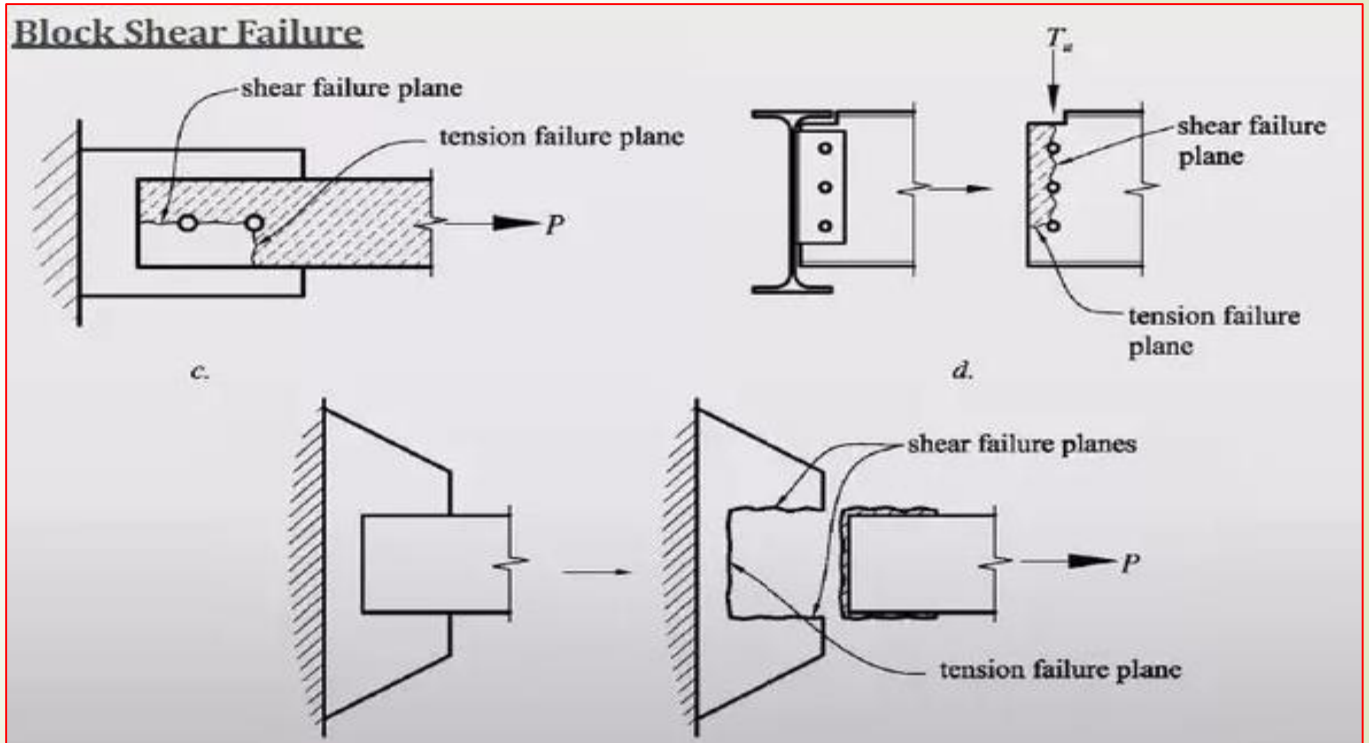
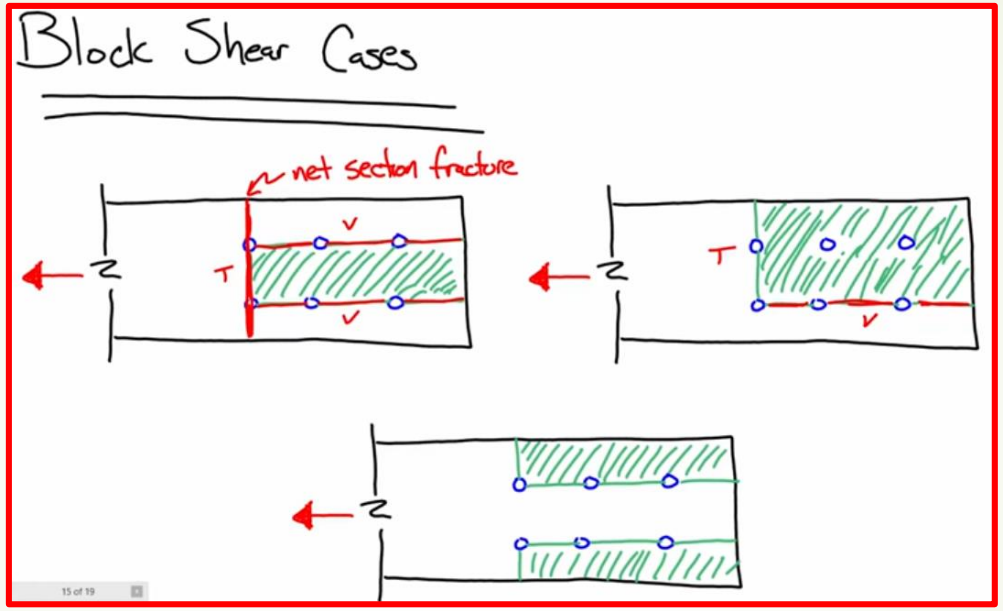


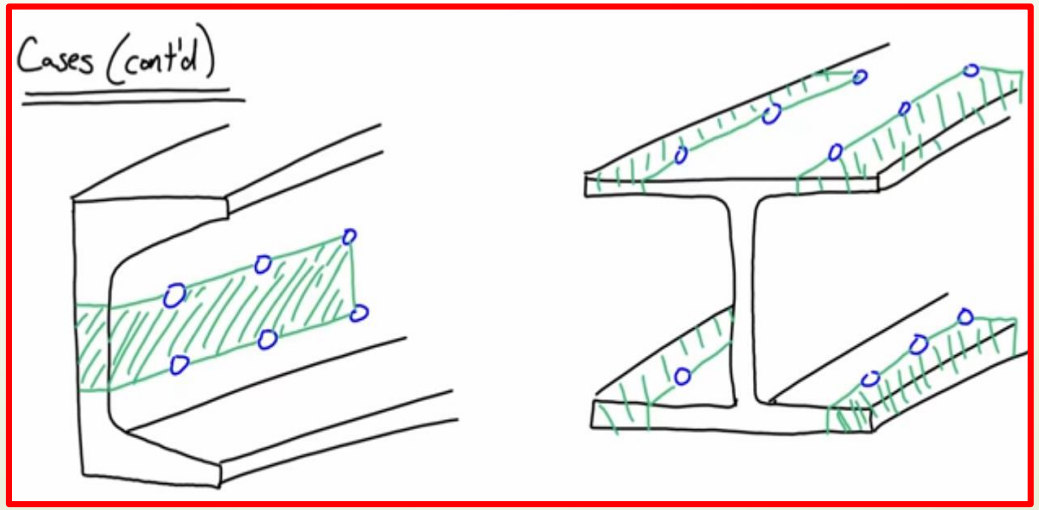
Figure 4.4 Block shear failure of single angle tension member





Limit State of Tension and Shear Rupture (Blok kırılma durumu)





Limit State of Tension and Shear Rupture (Blok kırılma durumu)

shear yielding of the element

$$R_n = 0.60F_yA_{gv}$$

where

A_{gv} = gross area subject to shear,

tensile rupture of elements

$U_{bs}F_uA_{nt}$

where

A_{nt} = net area subject to tension,

shear rupture of the element

$$R_n = 0.60F_uA_{nv}$$

where

A_{nv} = net area subject to shear,

13.4.3 – Blok Kırılma Dayanımı

Karakteristik blok kırılma dayanımı, R_n, kesme yüzeyi veya yüzeyleri boyunca akma ve kırılma sınır durumları ile çekme yüzeyi boyunca kopma sınır durumları esas alınarak, **Denk.(13.19)** ile hesaplanacaktır.

$$R_n = 0.60F_uA_{nv} + U_{bs}F_uA_{nt} \le 0.60F_yA_{gv} + U_{bs}F_uA_{nt} \qquad \textbf{(13.19)}$$

Tasarım blok kırılma dayanımı, ϕR_n (YDKT) veya *güvenli blok kırılma dayanımı*, R_n/Ω (GKT),

$$\phi = 0.75 \text{ (YDKT)} \quad \text{veya} \quad \Omega = 2.00 \text{ (GKT)}$$

alınarak belirlenecektir.

Buradaki terimler aşağıda açıklanmıştır.

A_{nt} : Çekme gerilmesi etkisindeki net alan.

U_{bs} : Çekme gerilmeleri yayılışını gözönüne alan bir katsayı.

Çekme gerilmeleri yayılışının üniform olduğu yüzeylerde, U_{bs} = 1.0, üniform olmadığı yüzeylerde ise U_{bs} = 0.5 olarak alınacaktır, (**Şekil 13.9**).

Where the tension stress is uniform, $U_{bs} = 1$; where the tension stress is nonuniform, $U_{bs} = 0.5$.

von Mises Failure Criterion (cont'd)

- When dealing with ductile materials (like steel), instead of using the Mohr-Coulomb failure criterion, we employ the von Mises failure criterion:
  - The 3D form is as follows:

$$F_y^2 \geq \frac{1}{2}\left[(\sigma_x-\sigma_y)^2+(\sigma_x-\sigma_z)^2+(\sigma_y-\sigma_z)^2+6(\tau_{xy}^2+\tau_{xz}^2+\tau_{yz}^2)\right]$$

- The reason this is important is for elements in pure shear.

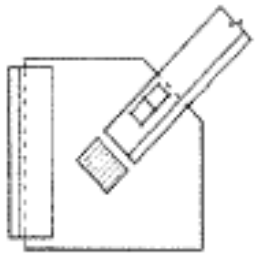
von Mises Failure Criterion (cont'd)

- For an element in pure shear, we'll set all other stress terms equal to zero:

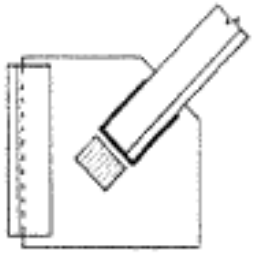
$$\begin{aligned} F_y^2 &\geq \frac{1}{2}\left[(\sigma_x-\sigma_y)^2+(\sigma_x-\sigma_z)^2+(\sigma_y-\sigma_z)^2+6(\tau_{xy}^2+\tau_{xz}^2+\tau_{yz}^2)\right] \\ &\geq \frac{1}{2}\left[(0-0)^2+(0-0)^2+(0-0)^2+6(\tau_{xy}^2+0^2+0^2)\right] \\ &\geq \frac{1}{2}\left[6(\tau_{xy}^2)\right] \\ &\geq 3\tau_{xy}^2 \end{aligned}$$

  - Therefore, an element in shear will yield when:

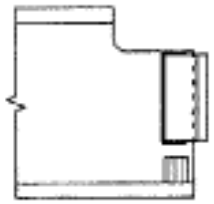
$$\tau_{\max} = \frac{F_y}{\sqrt{3}} \approx 0.577 F_y$$



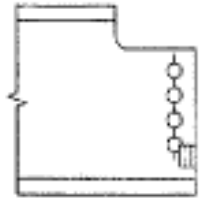
Korniyerin Bulonlu Birleşimi



Korniyerin Kaynaklı Birleşimi



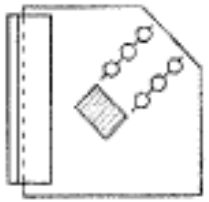
Korniyerin Kaynaklı Birleşimi



Tek Sıra Bulonlu Kiriş Uç Birleşimi

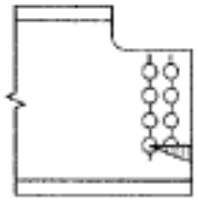


Korniyerin Bulonlu Uç Birleşimi



Düğüm Noktası Levhasına Bulonlu Birleşim

a)U_{bs} = 1.0 olan birleşim tipleri



Çok Sıra Bulonlu Kiriş Uç Birleşimi

b)U_{bs} = 0.5 olan birleşim tipi

Şekil 13.9 – Blok kırılma sınır durumunda çekme gerilmelerinin yayılışı

Limit State of Tension and Shear Rupture (Blok kırılma durumu)



Limit State of Tension and Shear Rupture (Blok kırılma durumu)

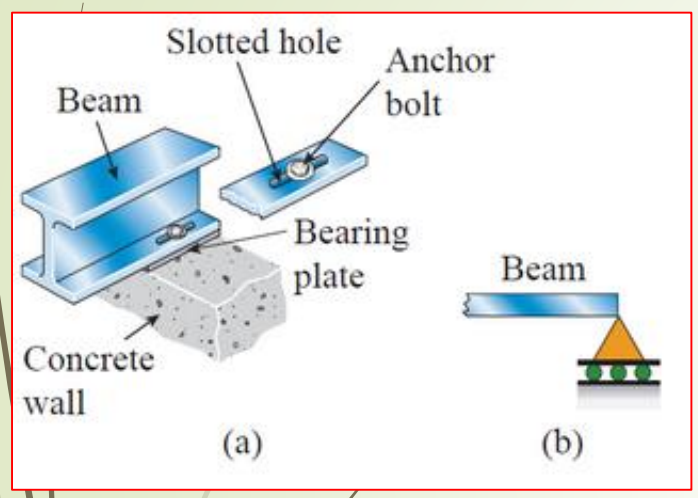


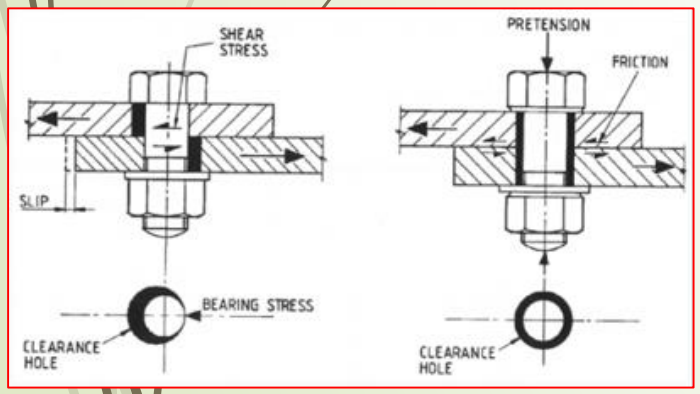
Delik çapları */Diameter of Hole*

TABLO 13.8 – KARAKTERİSTİK DELİK BOYUTLARI, (mm)

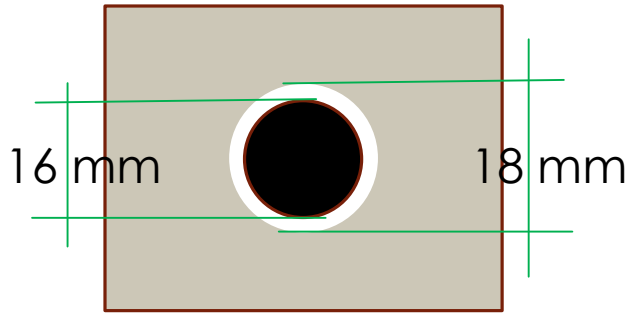
Bulon	Delik Boyutları			
	Standart Dairesel Delik Çapları	Büyük Dairesel Delik Çapları	Kısa Oval Delik (Genişlik × Uzunluk)	Uzun Oval Delik (Genişlik × Uzunluk)
M16	18	20	18 × 22	18 × 40
M20	22	24	22 × 26	22 × 50
M22	24	28	24 × 30	24 × 55
M24	26	30	26 × 32	26 × 60
M27	30	35	30 × 37	30 × 67
M30	33	38	33 × 40	33 × 75
≥ M36	$d+3$	$d+8$	$(d+3)\times(d+10)$	$(d+3)\times 2.5d$

In computing net area for tension and shear, the width of a bolt hole shall be taken as $^1/_{16}$ in. (2 mm) greater than the nominal dimension of the hole.

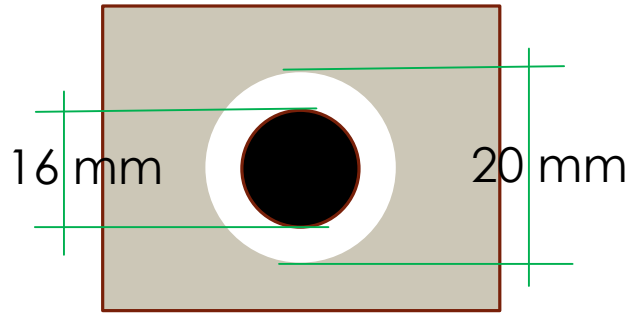




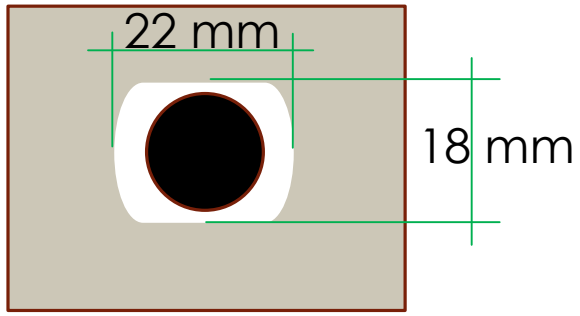
Standard hole



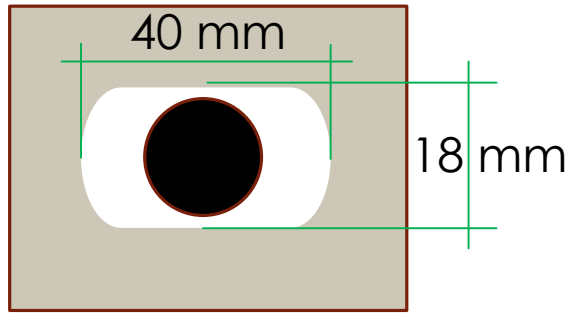
Over size hole



Short slot hole



Long slot hole



Bulon	Standart Dairesel Delik Çapları	Büyük Dairesel Delik Çapları	Kısa Oval Delik (Genişlik × Uzunluk)	Uzun Oval Delik (Genişlik × Uzunluk)
M16	18	20	18 × 22	18 × 40

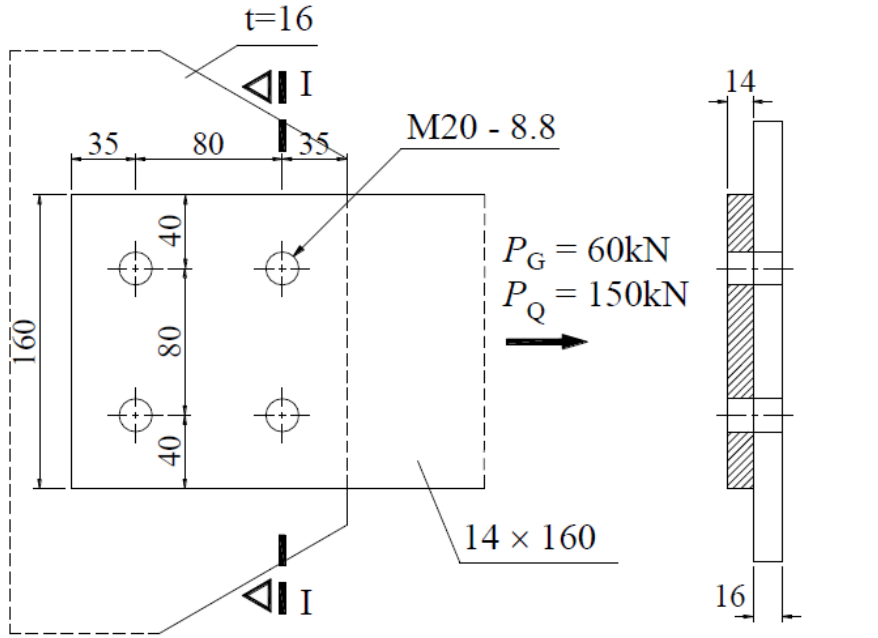
Örnek

Örnek 7. 1

Aşağıdaki şekilde bulonlu birleşim detayı verilen Levha 14×160 enkesitli 1.20m uzunluğundaki eleman, sabit ve hareketli yükler altında, sırasıyla, P_G = 60kN ve P_Q = 150kN eksenel çekme kuvvetleri etkisindedir.

a. Elemanın karakteristik çekme kuvveti dayanımının belirlenmesi

b. Elemanın tasarım çekme kuvveti dayanımının kontrolü (YDKT)



Enkesit

Levha 14×160	
$t = 14.0$ mm	$b = 160$ mm

Bulonlar

M20 – 8.8 (Standart dairesel delik çapı kullanılacaktır)

Çözüm

a. Elemanın karakteristik çekme kuvveti dayanımının belirlenmesi

Yönetmelik 7.2.1 uyarınca,

Akma sınır durumunda karakteristik çekme kuvveti dayanımı, T_n,

$$T_n = F_y A_g \tag{7.2}$$

$$T_n = F_y A_g = (235)(14\times160)10^{-3} = 526.40\ \text{kN}$$

Yönetmelik 7.2.2 uyarınca,

Kırılma sınır durumunda karakteristik çekme kuvveti dayanımı, T_n,

$$T_n = F_u A_e \tag{7.3}$$

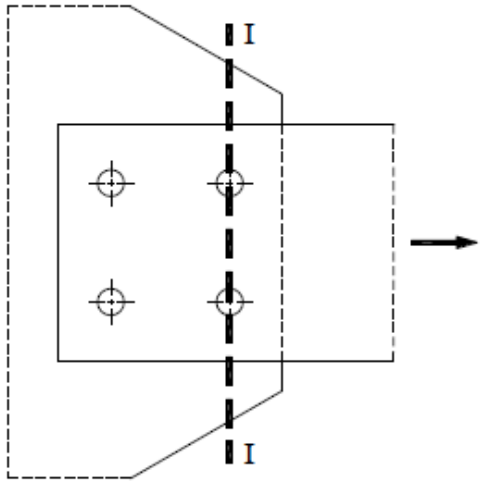
$$A_e = UA_n \tag{7.1}$$

Etkin net enkesit alanı, A_e,

Birleşim detayı incelendiğinde minimum net enkesit alanının bir adet kırılma çizgisi formu dikkate alınarak değerlendirilmesinin yeterli olduğu anlaşılmaktadır. Bu durum aşağıdaki şekilde gösterilmektedir.

$d_h = 20 + 2.0 = 22$ mm (*Yönetmelik* Tablo 13.8)

$d_e = 22 + 2.0 = 24$ mm (*Yönetmelik* 5.4.3(b))



$$A_n = A_g - \sum d_e t + \sum \frac{s^2 t}{4g} \qquad (5.3)$$

$A_n = (14)(160) - 2(24)(14) = 1568.0 \text{ mm}^2$

$U = 1.0$ (Tablo 7.1, Durum 1)

$$A_e = (1.0)(1568.0) = 1568.0\ \text{mm}^2$$

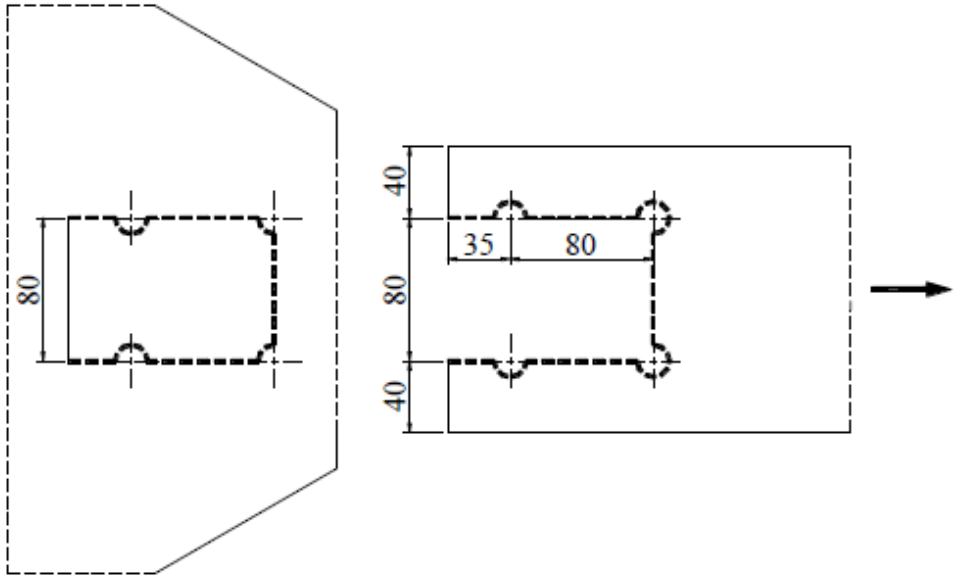
$$T_n = (360)(1568.0)10^{-3} = 564.48\ \text{kN}$$

Yönetmelik 7.2 ve 13.4.3 uyarınca,

Blok kırılma sınır durumunda karakteristik çekme kuvveti dayanımı, R_n,

Blok kırılma sınır durumunun değerlendirilmesinde aşağıdaki şekilde gösterilen olası göçme durumu öngörülebilir.

$$R_n = 0.60F_uA_{nv} + U_{bs}F_uA_{nt} \le 0.60F_yA_{gv} + U_{bs}F_uA_{nt} \qquad (13.19)$$



$A_{gv} = 2(35+80)(14) = 3220.0\ \text{mm}^2$

$A_{nv} = 3220.0 - 2(24+24/2)(14) = 2212.0\ \text{mm}^2$

$A_{nt} = (80)(14) - (24/2+24/2)(14) = 784.0\ \text{mm}^2$

$U_{bs} = 1.0$ (*Yönetmelik* 13.4.3)

$R_n = 0.60(360)(2212.0)10^{-3} + (1.0)(360)(784.0)10^{-3} = 760.03\ \text{kN}$

$0.60(235)(3220.0)10^{-3} + (1.0)(360)(784.0)10^{-3} = 736.26\ \text{kN}$

$R_n = 736.26\ \text{kN} < 760.03\ \text{kN}$

$R_n = 736.26\ \text{kN}$

YDKT	GKT
Gerekli çekme kuvveti dayanımı	
$P_u = 1.2P_G + 1.6P_Q$ $= 1.2(60) + 1.6(150) = 312\ \text{kN}$	$P_a = P_G + P_Q$ $= 60 + 150 = 210\ \text{kN}$
b. Elemanın tasarım çekme kuvveti dayanımının kontrolü	**c.** Elemanın güvenli çekme kuvveti dayanımının kontrolü
$T_d = \phi_t T_n = 0.90(526.40) = 473.76\ \text{kN}$ $T_d = \phi_t T_n = 0.75(564.48) = 423.36\ \text{kN}$ $T_d = \phi R_n = 0.75(736.26) = 552.20\ \text{kN}$	$T_g = T_n/\Omega_t = 526.40/1.67 = 315.21\ \text{kN}$ $T_g = T_n/\Omega_t = 564.48/2.00 = 282.24\ \text{kN}$ $T_g = R_n/\Omega = 736.26/2.00 = 368.13\ \text{kN}$
$\frac{P_u}{T_d} = \frac{312}{423.36} = 0.74 \le 1.0$ ✓	$\frac{P_a}{T_g} = \frac{210}{282.24} = 0.74 \le 1.0$ ✓

Çekme elemanı narinlik oranı kontrolü, *Yönetmelik* 7.1.1 uyarınca, $L/i \le 300$

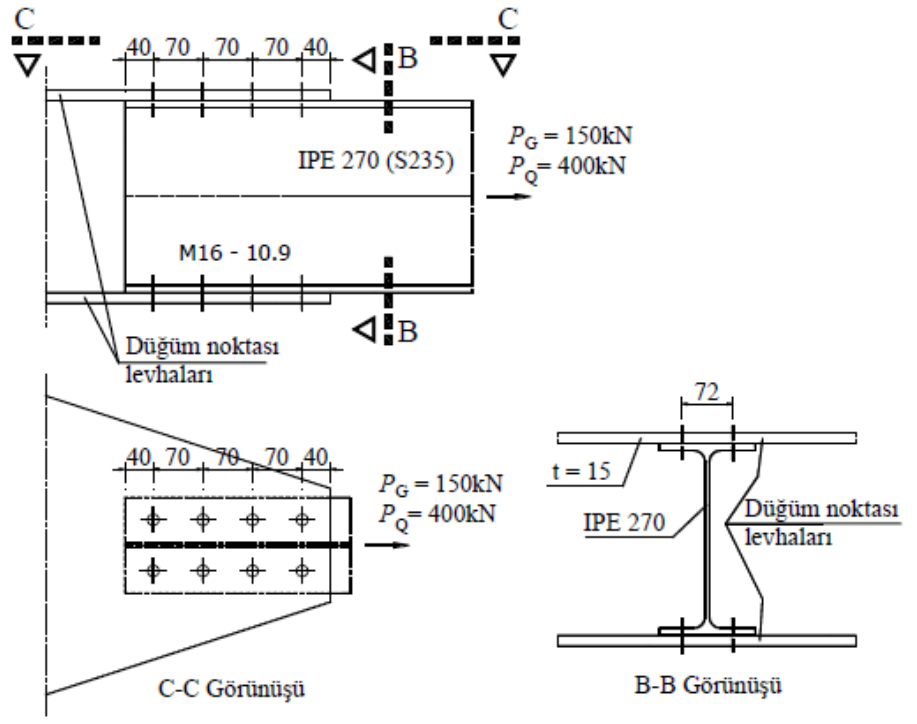
$i_{min} \cong 0.2887t \cong 4.04\ \text{mm}$ alınarak, $1200/4.04 = 297 \le 300$ ✓

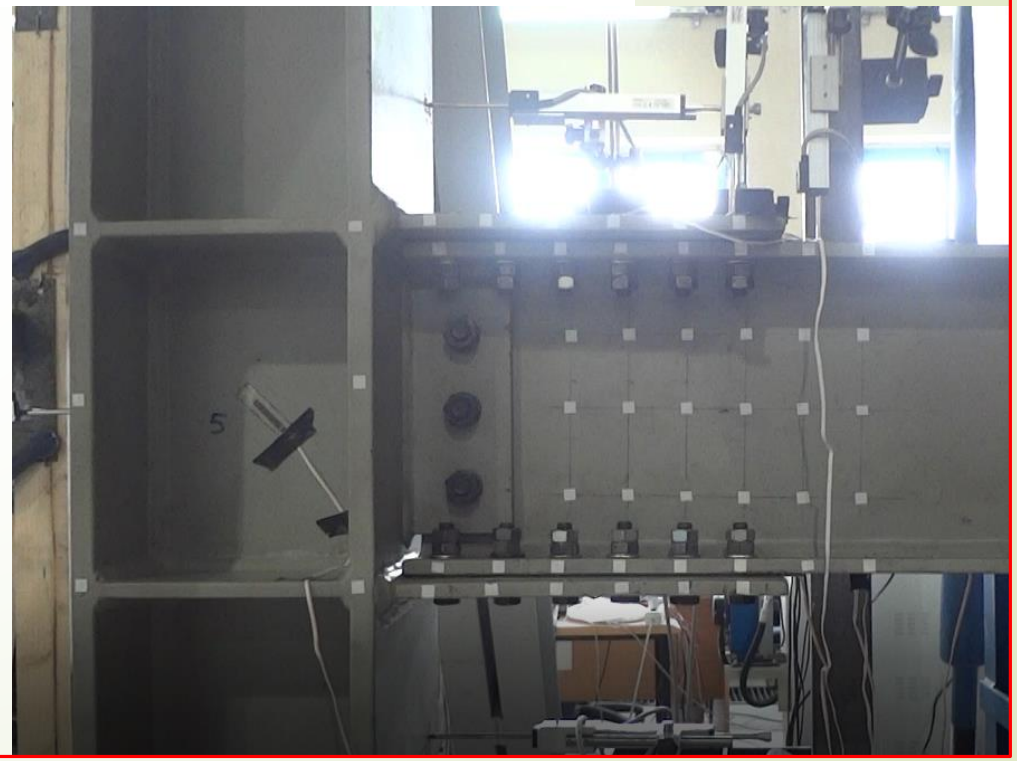
Örnek 7.2

Aşağıdaki şekilde bulonlu birleşim detayı verilen 5.0m uzunluğundaki IPE 270 profili ile boyutlandırılan eleman, sabit ve hareketli yükler altında, sırasıyla, P_G = 150kN ve P_Q = 400kN eksenel çekme kuvvetleri etkisindedir.

a. Elemanın karakteristik çekme kuvveti dayanımının belirlenmesi

b. Elemanın tasarım çekme kuvveti dayanımının kontrolü (YDKT)





Çelik sınıfı

S 235 $F_y = 235$ N/mm^2 $F_u = 360$ N/mm^2 (*Yönetmelik* Tablo 2.1A)

Enkesit

IPE 270				
$A_g = 4590$ mm^2	$d = 270$ mm	$b_f = 135$ mm	$t_f = 10.2$ mm	$i_y = 30.2$ mm

Bulonlar

M16 – 10.9 (Standart dairesel delik çapı kullanılacaktır)

Çözüm

a. Elemanın karakteristik çekme kuvveti dayanımının belirlenmesi

Yönetmelik 7.2.1 uyarınca,

Akma sınır durumunda karakteristik çekme kuvveti dayanımı, T_n,

$$T_n = F_y A_g \qquad (7.2)$$

$$T_n = F_y A_g = (235)(4590)10^{-3} = 1078.65 \text{ kN}$$

Yönetmelik 7.2.2 uyarınca,

Kırılma sınır durumunda karakteristik çekme kuvveti dayanımı, T_n,

$$T_n = F_u A_e \qquad (7.3)$$

$$A_e = UA_n \qquad (7.1)$$

Etkin net enkesit alanı, A_e,

$d_h = 16 + 2.0 = 18$ mm (*Yönetmelik* Tablo 13.8)

$d_e = 18 + 2.0 = 20$ mm (*Yönetmelik* 5.4.3(b))

$$A_n = A_g - 4d_e t_f$$

$$A_n = 4590 - 4(20)(10.2) = 3774.0 \text{ mm}^2$$

$U = 1 - \frac{\overline{x}}{l}$ (Tablo 7.1, Durum 2) $\overline{x} = y_g = 29.7$ mm $l = 3(70) = 210$ mm

Sadece başlıklarından bağlanan I-enkesitli elemanlarda, $\overline{x}$, yarım I-enkesitin (burada ½ IPE 270 profilinin) ağırlık merkezinin birleşim yüzeyine uzaklığı olarak tanımlanır.

$$U = 1 - \frac{29.7}{210} = 0.859 \geq U_{min} = \frac{2b_f t_f}{A_g} = \frac{2(135)(10.2)}{4590} = 0.60$$

Not: Gerilme düzensizliği etki katsayısının minimum değeri, U_{min}, birleşen enkesit parçalarının toplam enkesit alanının eleman kayıpsız enkesit alanına oranı olarak hesaplanacaktır.

$$A_e = (0.859)(3774.0) = 3241.87 \text{ mm}^2$$

$$T_n = (360)(3241.87)10^{-3} = 1167.07 \text{ kN}$$

Yönetmelik 7.2 ve 13.4.3 uyarınca,

Blok kırılma sınır durumunda karakteristik çekme kuvveti dayanımı, R_n,

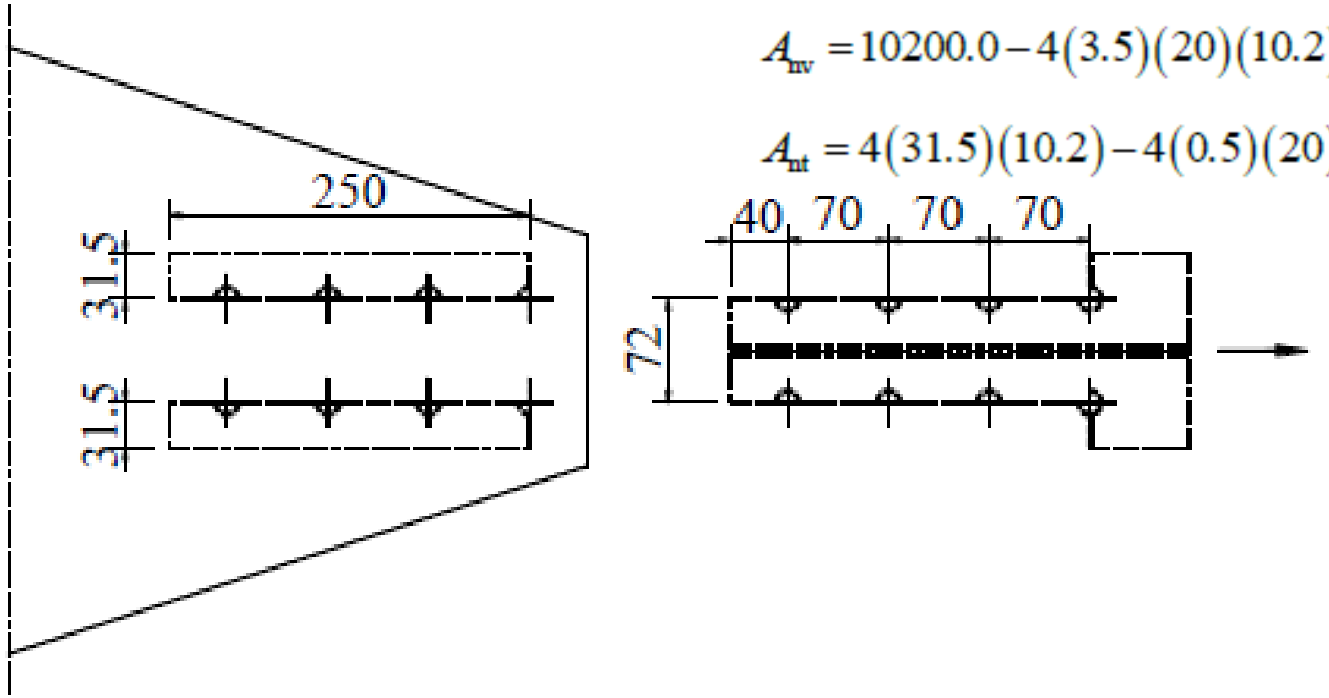
Blok kırılma sınır durumunun değerlendirilmesinde aşağıdaki şekilde gösterilen olası göçme durumu öngörülebilir.

$$R_n = 0.60F_u A_{nv} + U_{bs}F_u A_{nt} \le 0.60F_y A_{gv} + U_{bs}F_u A_{nt} \qquad (13.19)$$

$$A_{gv} = 4(40+70+70+70)(10.2) = 10200.0\ \text{mm}^2$$

$$A_{nv} = 10200.0 - 4(3.5)(20)(10.2) = 7344.0\ \text{mm}^2$$

$$A_{nt} = 4(31.5)(10.2) - 4(0.5)(20)(10.2) = 877.20\ \text{mm}^2$$



$U_{bs} = 1.0$

$R_n = 0.60(360)(7344.0)10^{-3} + (1.0)(360)(877.20)10^{-3} = 1902.10\ \text{kN}$

$0.60(235)(10200.0)10^{-3} + (1.0)(360)(877.20)10^{-3} = 1753.99\ \text{kN}$

$R_n = 1902.10\ \text{kN} > 1753.99\ \text{kN}$

$R_n = 1753.99\ \text{kN}$

YDKT	GKT
Gerekli çekme kuvveti dayanımı	
$P_u = 1.2P_G + 1.6P_Q$ $= 1.2(150) + 1.6(400) = 820\ \text{kN}$	$P_a = P_G + P_Q$ $= 150 + 400 = 550\ \text{kN}$
b. Elemanın tasarım çekme kuvveti dayanımının kontrolü	**c.** Elemanın güvenli çekme kuvveti dayanımının kontrolü
$T_d = \phi_t T_n = 0.90(1078.65) = 970.79\ \text{kN}$ $T_d = \phi_t T_n = 0.75(1167.07) = 875.30\ \text{kN}$ $T_d = \phi R_n = 0.75(1753.99) = 1315.49\ \text{kN}$	$T_g = T_n/\Omega_t = 1078.65/1.67 = 645.90\ \text{kN}$ $T_g = T_n/\Omega_t = 1167.07/2.00 = 583.54\ \text{kN}$ $T_g = R_n/\Omega = 1753.99/2.00 = 877.00\ \text{kN}$
$\frac{P_u}{T_d} = \frac{820}{875.30} = 0.94 \le 1.0$ ✓	$\frac{P_a}{T_g} = \frac{550}{583.54} = 0.94 \le 1.0$ ✓

Yönetmelik 7.1.1 uyarınca,

Çekme elemanı narinlik oranı kontrolü, $L/i \le 300$

$i_{min} = i_y = 30.2$ mm alınarak, $5000/30.2 = 166 \le 300$ ✓

STAGGERED FASTENERS

If a tension member connection is made with bolts, the net area will be maximized if the fasteners are placed in a single line. Sometimes space limitations, such as a limit on dimension *a* in Figure 3.14a, necessitate using more than one line. If so, the reduction in cross-sectional area is minimized if the fasteners are arranged in a staggered pattern, as shown. Sometimes staggered fasteners are required by the geometry of a connection, such as the one shown in Figure 3.14b. In either case, any cross section passing through holes will pass through fewer holes than if the fasteners are not staggered.

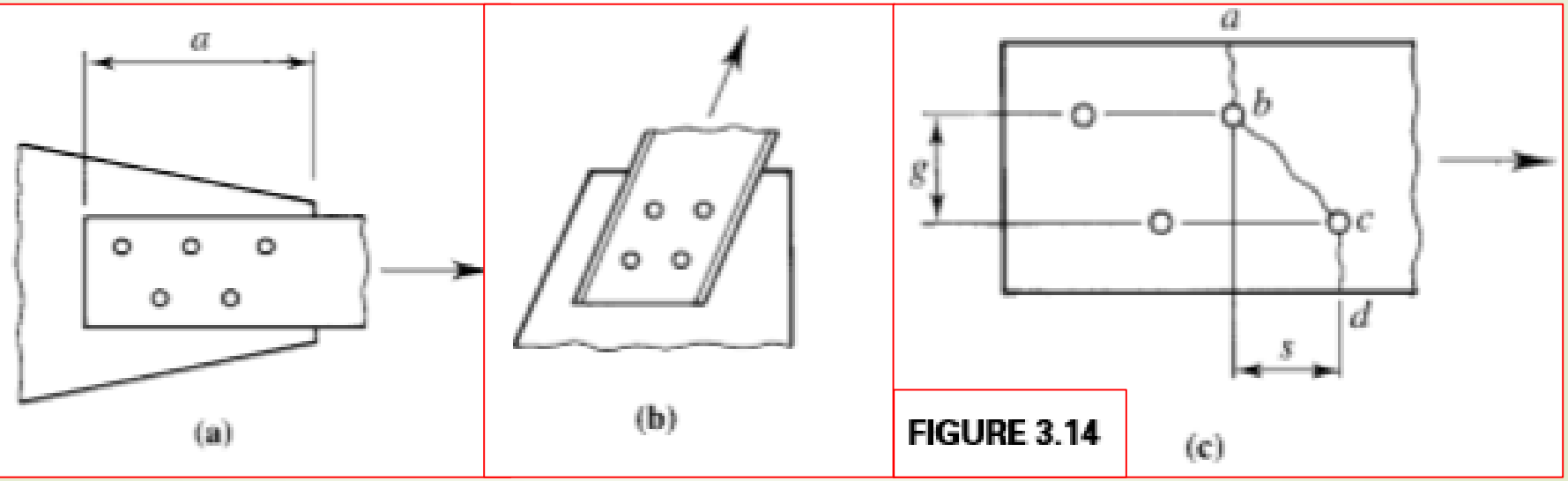


FIGURE 3.14

$$A_n = A_g - \sum d_e t + \sum \frac{s^2 t}{4g}$$

d_e : Etkin delik çapı,(= d_h + 2mm).

t : Eleman kalınlığı.

For a chain of holes extending across a part in any diagonal or zigzag line, the net width of the part shall be obtained by deducting from the gross width the sum of the diameters or slot dimensions as provided in this section, of all holes in the chain, and adding, for each gage space in the chain, the quantity $s^2/4g$,

where

g = transverse center-to-center spacing (gage) between fastener gage lines, in. (mm)

s = longitudinal center-to-center spacing (pitch) of any two consecutive holes, in. (mm)

For angles, the gage for holes in opposite adjacent legs shall be the sum of the gages from the back of the angles less the thickness.

For slotted HSS welded to a gusset plate, the net area, A_n, is the gross area minus the product of the thickness and the total width of material that is removed to form the slot.

EXAMPLE 3.7

An angle with staggered fasteners in each leg is shown in Figure 3.17. A36 steel is used, and holes are for ⅞-inch-diameter bolts.

a. Determine the design strength for LRFD.

b. Determine the allowable strength for ASD.

SOLUTION

From the dimensions and properties tables, the gross area is $A_g = 6.80$ in.^{2}. The effective hole diameter is ⅞ + ⅛ = 1 in.

For line *abdf*, the net area is

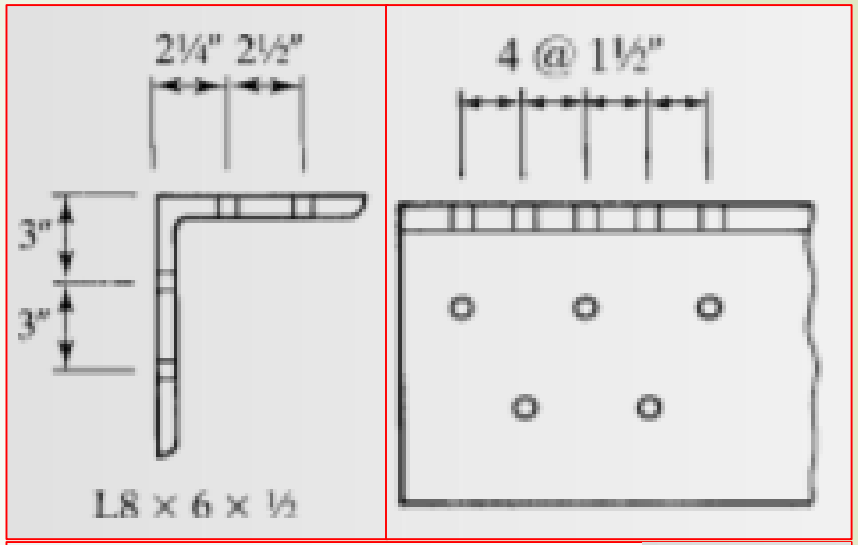
$$A_n = A_g - \sum t_w \times (d \text{ or } d')$$

$$= 6.80 - 0.5(1.0) \times 2 = 5.80 \text{ in.}^2$$

For line *abceg*,

$$A_n = 6.80 - 0.5(1.0) - 0.5\left[1.0 - \frac{(1.5)^2}{4(2.5)}\right] - 0.5(1.0) = 5.413 \text{ in.}^2$$

Because ⅒ of the load has been transferred from the member by the fastener at *d*, this potential failure line must resist only ⅘ of the load. Therefore, the net area



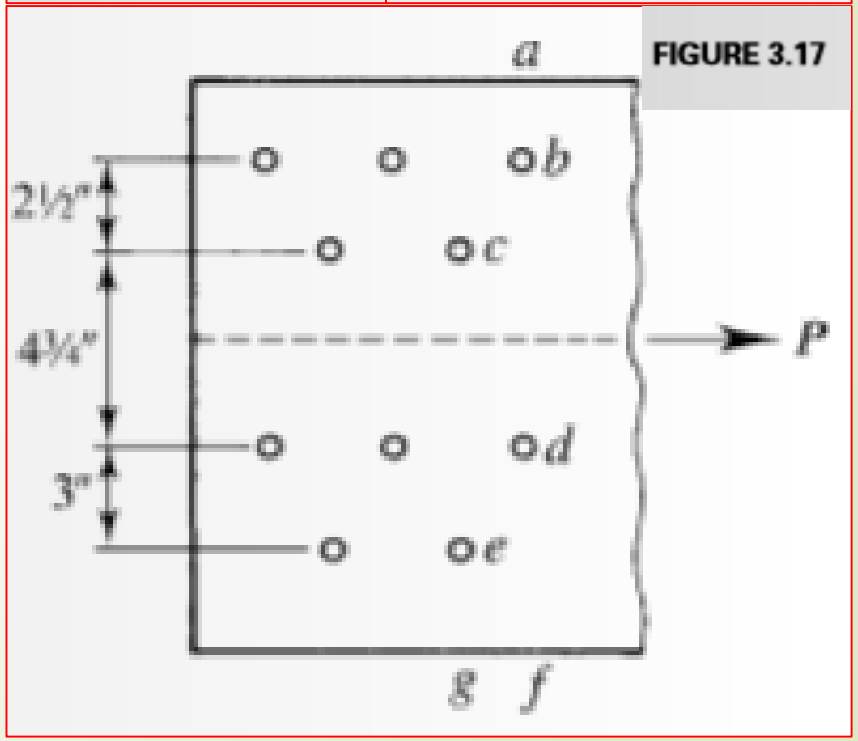


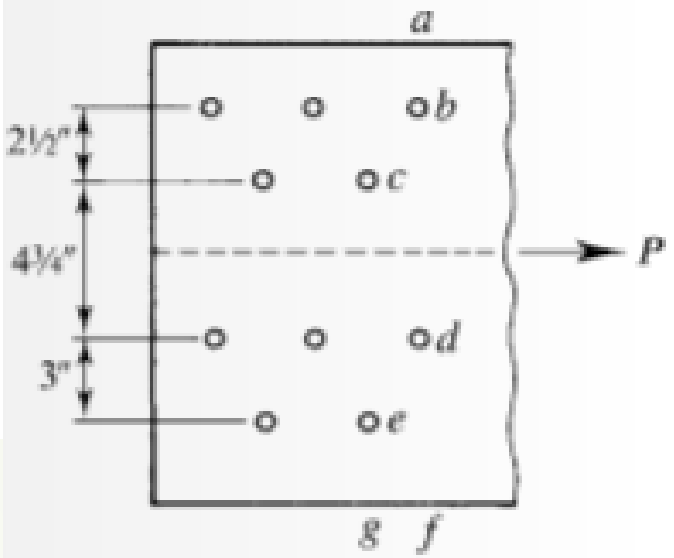
FIGURE 3.17

of 5.413 in.^{2} should be multiplied by 10% to obtain a net area that can be compared with those lines that resist the full load. Use $A_n = 5.413(10\%) = 6.014$ in.^{2} For line *abcdeg*,

$$g_{ad} = 3 + 2.25 - 0.5 = 4.75 \text{ in.}$$

$$A_n = 6.80 - 0.5(1.0) - 0.5\left[1.0 - \frac{(1.5)^2}{4(2.5)}\right] - 0.5\left[1.0 - \frac{(1.5)^2}{4(4.75)}\right] - 0.5\left[1.0 - \frac{(1.5)^2}{4(3)}\right]$$

$$= 5.065 \text{ in.}^2$$



The last case controls; use

$$A_n = 5.065 \text{ in.}^2$$

Both legs of the angle are connected, so

$$A_e = A_n = 5.065 \text{ in.}^2$$

The nominal strength based on fracture is

$$P_n = F_u A_e = 58(5.065) = 293.8 \text{ kips}$$

The nominal strength based on yielding is

$$P_n = F_y A_g = 36(6.80) = 244.8 \text{ kips}$$

a. The design strength based on fracture is

$$\phi_t P_n = 0.75(293.8) = 220 \text{ kips}$$

The design strength based on yielding is

$$\phi_t P_n = 0.90(244.8) = 220 \text{ kips}$$

ANSWER Design strength = 220 kips.

b. For the limit state of fracture, the allowable stress is

$$F_t = 0.5F_u = 0.5(58) = 29.0 \text{ ksi}$$

and the allowable strength is

$$F_t A_e = 29.0(5.065) = 147 \text{ kips}$$

For yielding,

$$F_t = 0.6F_y = 0.6(36) = 21.6 \text{ ksi}$$

$$F_t A_g = 21.6(6.80) = 147 \text{ kips}$$

ANSWER Allowable strength = 147 kips.

Slenderness Limitations

- In addition, we need to discuss service limits for tension members.
  - Specifically, a limit on the slenderness of tension members.
  - Then again, do we?

D1. SLENDERNESS LIMITATIONS

There is no maximum slenderness limit for members in tension.

User Note: For members designed on the basis of tension, the slenderness ratio *L/r* preferably should not exceed 300. This suggestion does not apply to rods or hangers in tension.

These slenderness limitations are in place to prevent:

- Excessive sag
- Excessive vibrations
- Damage during shipping
- Damage during erection
- Excessive deflection under the members own weight

For most cases in this course, we will employ them.
Other than threaded rods (we'll discuss those later).

$$\frac{L}{r_{min}} \leq 300$$

L = member length

r_{min} = minimum radius of gyration

- Use minimum of r_x and r_y for symmetric shapes (such as W-shapes)
- Use minimum "r" with respect to principal axes for unsymmetric shapes (like angles)